Algebra 1: Test 2 Notes

Lilian

October 2026

Linear Operators

1 EIGENANALYSIS	SECTION 1.1 The key question	2
	SECTION 1.2 Eigenvectors and the characteristic polynomial	2
	SECTION 1.3 Independence of eigenspaces and the diagonalizability test	4
	SECTION 1.4 Other proofs	6
2 TRIANGULAR FORM AND NILPOTENT OPERATORS	SECTION 2.1 Invariant subspaces and restrictions	10
	SECTION 2.2 Triangular form	11
	SECTION 2.3 Nilpotent operators	12
	SECTION 2.4 Other proofs	14
3 CAYLEY--HAMILTON, MINIMAL POLYNOMIALS, TOOLS	SECTION 3.1 Projections	18
	SECTION 3.2 Invariant subspaces of diagonalizable operators	19
	SECTION 3.3 The Cayley–Hamilton theorem	20
	SECTION 3.4 The minimal polynomial	22
	SECTION 3.5 Block matrices	23
	SECTION 3.6 Operators on spaces of matrices	24
4 FITTING DECOMPOSITION AND JORDAN FORM	SECTION 4.1 Fitting decomposition	28
	SECTION 4.2 Generalized eigenspaces	29
	SECTION 4.3 Nilpotent Jordan form (Step 2)	31
	SECTION 4.4 Uniqueness (Step 3)	35

Inner Product Spaces

5 EUCLIDEAN SPACES	SECTION 5.1 The dot product and inner products	42
	SECTION 5.2 Orthonormal bases	43
	SECTION 5.3 Orthogonal complements	43
	SECTION 5.4 Projection onto a line, Cauchy–Schwarz	45

The Test

5 EUCLIDEAN SPACES	SECTION 5.5 The Gram–Schmidt process	46
6 HERMITIAN SPACES	SECTION 6.1 Complex inner products	52
	SECTION 6.2 What carries over	53
	SECTION 6.3 Other proofs	53
7 ORTHOGONAL AND UNITARY OPERATORS	SECTION 7.1 Definition and basic properties	55
	SECTION 7.2 The orthogonal groups	56
	SECTION 7.3 Euler's theorem on rotations of $\mathbb{R}^3$	57
8 WRITING AND STANDARD METHODS	SECTION 8.1 Order of work	63
	SECTION 8.2 Rules for the answers	63
	SECTION 8.3 Standard computations	64
9 MOCK TEST	CHAPTER 9 Mock Test	66
A FIELDS AND COUNTING OVER $\mathbb{F}_p$	CHAPTER A Fields and Counting over $\mathbb{F}_p$	68

About these notes

These notes rewrite the board work from Sep 28 to Oct 22, 2026, starting at eigenanalysis, and add alternative proofs and problems from the books. They cover the syllabus of Test 2 (Friday Oct 30, 2–6 pm), and in places go beyond it.

Proofs. Proofs are outlines. Each step marked (*Ex*) is for you to fill in. Where a result has several proofs, the one from class comes first and the others follow in a section called "Other proofs".

Exercises. Every chapter ends with exercises in three groups: from the board and the tutorials, from the books, and harder or unusual problems. The source is given after the exercise number. Each exercise carries a rank:

[1] before the test, first: the syllabus core, close to what the instructor sets;

[2] before the test, if there is time;

[3] after the test: deepens the theory or covers side topics;

[4] challenge problems, some beyond the syllabus.

Section 8.1 lists the rank-1 exercises in the recommended order.

Sources. Artin, *Algebra*, 2nd ed. ("Artin 4.7.3" is Chapter 4, Section 7, Exercise 3; "Artin 5.M.3" is a miscellaneous problem of Chapter 5). Axler, *Linear Algebra Done Right*, 4th ed. ("Axler 8A.13"), open access under CC BY-NC 4.0; its exercises are restated here, sometimes adapted. Halmos, *Finite-Dimensional Vector Spaces* ("Halmos §58.9"). Evan Chen, *An Infinitely Large Napkin* ("Napkin 13C"), CC BY-SA 4.0. Artin's and Halmos's problems are restated in my own words.

Conventions. Inner products here are conjugate-linear in the *first* slot and linear in the second, as in class. Axler and Halmos use the opposite convention, so convert their formulas first. Unless stated otherwise, T is an operator on a finite-dimensional vector space V of dimension n over a field $\mathbb{F}$.

Out of the syllabus.

- The proofs of the Fitting decomposition, of Step 1 (generalized eigenspace decomposition) and of Step 2 (Jordan basis for a nilpotent operator). Their statements are needed. The proofs are included here anyway, since they were done in class.
- Symmetric and Hermitian operators, and every version of the spectral theorem, statements included (the Oct 21 boards 3–4 and the Oct 22 boards 3(2), 4 bottom and 5).

A rule from the 2025 papers. The 2025 Test 2 and Endsem both began: "Do not use the Jordan form unless the question mentions it." Practise diagonalizability and similarity arguments with eigenspace dimensions, kernel dimensions of powers, and characteristic and minimal polynomials.

Part 1



Linear Operators

CHAPTER 1	Eigenanalysis	2
CHAPTER 2	Triangular Form and Nilpotent Operators	10
CHAPTER 3	Cayley–Hamilton, Minimal Polynomials, Tools	18
CHAPTER 4	Fitting Decomposition and Jordan Form	28

1 Eigenanalysis

1.1 The key question

The key question for an operator $T: V \to V$ is how to find a "good basis" of V, one in which the matrix of T is as simple as possible. "Simple" means diagonal, if possible. This is often possible, but not always. Three subquestions:

1. When can T be diagonalized?
2. When it can, how do we find a magic basis?
3. When it cannot, what can we do?

Definition 1.1

T is *diagonalizable* if there is a basis $B = (v_1, \ldots, v_n)$ of V with $T(v_i) = \lambda_i v_i$ for each i. Then the matrix of T with respect to B is $\operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.

For $V = \mathbb{F}^n$ and $T(x) = Ax$, diagonalizing A means finding an invertible P with $P^{-1}AP = D$ diagonal. Write $P = [v_1 \cdots v_n]$ by columns. The i-th column of AP is Av_i, and the i-th column of PD is $\lambda_i v_i$, so

$$P^{-1}AP = D \iff AP = PD \iff Av_i = \lambda_i v_i \text{ for all } i.$$

So the columns of P form a basis of eigenvectors, an *eigenbasis*, and D lists the matching eigenvalues.

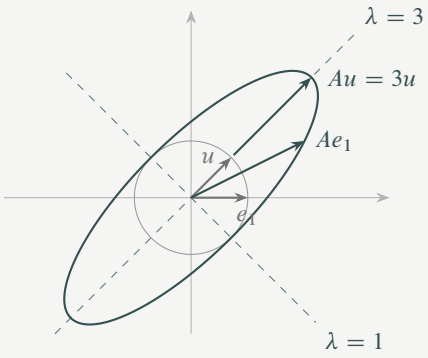
1.2 Eigenvectors and the characteristic polynomial

Definition 1.2

A nonzero $v \in V$ is an *eigenvector* of T if $Tv = \lambda v$ for some $\lambda \in \mathbb{F}$. Then λ is the *eigenvalue* corresponding to v, and v is a λ-eigenvector. For $\lambda \in \mathbb{F}$, the *λ-eigenspace* of T is $V_\lambda = \ker(\lambda I - T)$: the λ-eigenvectors together with 0.

Example 1.1 **What an eigenvector looks like**

Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Then $A(1,1)^T = 3(1,1)^T$ and $A(1,-1)^T = (1,-1)^T$, so the lines $y = x$ and $y = -x$ are mapped into themselves, stretched by 3 and by 1. A vector off these lines changes direction: $Ae_1 = (2,1)^T$. The figure shows the unit circle and its image, an ellipse whose axes are the two eigenlines.



Since $Tv = \lambda v \iff (\lambda I - T)v = 0$,

λ is an eigenvalue $\iff \ker(\lambda I - T) \neq 0 \iff \lambda I - T$ is not invertible $\iff \det(\lambda I - T) = 0$.

So we find the eigenvalues first, by treating λ as an unknown, computing $\det(\lambda I - T)$ and solving for λ. Then, for each eigenvalue, we solve $(\lambda I - T)v = 0$ by row reduction to get a basis of the eigenspace.

Definition 1.3

The *characteristic polynomial* of T is $p_T(x) = \det(xI - A)$, where A is the matrix of T in some basis. The *algebraic multiplicity* of an eigenvalue λ is its multiplicity as a root of p_T; its *geometric multiplicity* is $\dim V_\lambda$.

Proposition 1.1. *p_T does not depend on the basis used to compute it, and*

$$p_T(x) = x^n - \operatorname{tr}(T)\, x^{n-1} + \cdots + (-1)^n \det T.$$

Proof. If A is the matrix in one basis, the matrix in another basis is $P^{-1}AP$ for the change-of-basis matrix P. Since $xI = P^{-1}(xI)P$,

$$\det\big(xI - P^{-1}AP\big) = \det\left(P^{-1}(xI - A)P\right) = \det(P)^{-1}\det(xI - A)\det(P) = \det(xI - A).$$

For the coefficients: in the expansion of $\det(xI - A)$ as a sum over permutations, only the product of the diagonal entries $\prod(x - a_{ii})$ contributes to x^n and x^{n-1}, since any other permutation misses at least two diagonal entries; this gives $x^n - (\sum a_{ii})x^{n-1}$. The constant term is $p_T(0) = \det(-A) = (-1)^n \det A$. □

Example 1.2 **Fibonacci matrix**

$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ has $p_A(x) = x^2 - x - 1$, with roots $\lambda_{1,2} = \frac{1 \pm \sqrt{5}}{2}$. Taking eigenvectors $v_1 \in \ker(A - \lambda_1 I)$ and $v_2 \in \ker(A - \lambda_2 I)$ gives an eigenbasis, and diagonalizing A gives the closed formula for the Fibonacci numbers.

Example 1.3 **Rotation by $\pi/2$**

$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ has $p_R(x) = x^2 + 1$, which has no root in $\mathbb{R}$, as expected: no line is mapped to itself. If we regard $T(v) = Rv$ on $\mathbb{C}^2$, then T has eigenvalues $\pm i$; for example $R \begin{bmatrix} 1 \\ i \end{bmatrix} = -i \begin{bmatrix} 1 \\ i \end{bmatrix}$. Eigenvectors with distinct eigenvalues are not multiples of each other, so the two eigenvectors form a basis of $\mathbb{C}^2$, and T is diagonalizable over $\mathbb{C}$.

Example 1.4 **Not diagonalizable**

$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has 0 as its only eigenvalue. If N were diagonalizable, then $P^{-1}NP = \operatorname{diag}(0,0) = 0$, so $N = P0P^{-1} = 0$, a contradiction.

Remark 1.2.1. The eigenvalues of a triangular matrix are its diagonal entries $a_{11}, \ldots, a_{nn}$, because $xI - A$ is triangular too and so $p_A(x) = \prod(x - a_{ii})$.

1.3 Independence of eigenspaces and the diagonalizability test

Theorem 1.2

Let $Tv_i = \lambda_i v_i$ for $i = 1, \ldots, k$, with $\lambda_1, \ldots, \lambda_k$ distinct. If $v_1 + \cdots + v_k = 0$, then every $v_i = 0$. In other words, eigenspaces for distinct eigenvalues are independent, so the sum of the eigenspaces of any operator is direct; in particular, eigenvectors with distinct eigenvalues are linearly independent.

Proof (class). Induction on k; for $k = 1$ there is nothing to prove. Suppose the statement holds for $k - 1$ vectors, and let $v_1 + \cdots + v_k = 0$. Apply T, and separately

multiply by λ_1:

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_k v_k = 0, \qquad \lambda_1 v_1 + \lambda_1 v_2 + \cdots + \lambda_1 v_k = 0.$$

Subtracting removes v_1: $(\lambda_2 - \lambda_1)v_2 + \cdots + (\lambda_k - \lambda_1)v_k = 0$. Each $(\lambda_j - \lambda_1)v_j$ is again in V_{λ_j}, so by induction every $(\lambda_j - \lambda_1)v_j = 0$, and since $\lambda_j \neq \lambda_1$, every $v_j = 0$ for $j \geq 2$. Then $v_1 = -(v_2 + \cdots + v_k) = 0$. □

Corollary 1.3. *The following are equivalent:*

1. *T is diagonalizable;*
2. *$V = \bigoplus_\lambda V_\lambda$, the sum over the distinct eigenvalues;*
3. *$\sum_\lambda \dim V_\lambda = \dim V$;*
4. *p_T factors fully into linear factors over $\mathbb{F}$, and the geometric multiplicity of every eigenvalue equals its algebraic multiplicity.*

In particular, if T has n distinct eigenvalues, then it is diagonalizable, and each eigenspace is 1-dimensional. The converse is false: take $T = I$ or $T = 0$.

Proof. By theorem 1.2 the sum $\sum_\lambda V_\lambda$ is always direct, so its dimension is $\sum_\lambda \dim V_\lambda$. Hence (b) $\iff$ (c). An eigenbasis can be grouped by eigenvalue, giving bases of the V_λ that together span V; conversely, putting together bases of the V_λ gives an eigenbasis when the sum is V. Hence (a) $\iff$ (b). For (c) $\iff$ (d), let $g_\lambda = \dim V_\lambda$ and m_λ the algebraic multiplicity. Always $g_\lambda \leq m_\lambda$ (proposition 1.4 below) and $\sum_\lambda m_\lambda \leq n = \deg p_T$, with equality exactly when p_T factors fully. So $\sum g_\lambda = n$ holds if and only if both $\sum m_\lambda = n$ and every $g_\lambda = m_\lambda$ (*Ex*). □

So the recipe to test diagonalizability is: (1) find $p_T(x) = \det(xI - T)$ and its roots; (2) for each root λ, find a basis of $\ker(\lambda I - T)$. The test can fail for two reasons:

(i) *too few eigenvalues*: p_T does not factor fully over $\mathbb{F}$, which is a defect of the field;

(ii) *too few eigenvectors for some eigenvalue*: $\dim V_\lambda < m_\lambda$.

From now on we mostly assume away (i), that is, p_T factors fully, and deal with (ii).

Proposition 1.4 (geometric ≤ algebraic multiplicity)**.** *For every eigenvalue λ, $\dim V_\lambda \leq m_\lambda$.*

Proof. Let $g = \dim V_\lambda$, take a basis $v_1, \ldots, v_g$ of V_λ and extend it to a basis of V. Since $Tv_i = \lambda v_i$ for $i \leq g$, the first g columns of the matrix are $\lambda e_1, \ldots, \lambda e_g$, so

$$[T] = \begin{bmatrix} \lambda I_g & B \\ 0 & D \end{bmatrix}, \qquad xI - [T] = \begin{bmatrix} (x-\lambda) I_g & -B \\ 0 & xI - D \end{bmatrix}.$$

The determinant of a block triangular matrix is the product of the determinants of its diagonal blocks (proposition 3.6), so $p_T(x) = (x-\lambda)^g \, p_D(x)$. Hence $(x-\lambda)^g$ divides p_T, that is, $g \leq m_\lambda$. □

1.4 Other proofs

Theorem 1.2 by a Vandermonde determinant. Suppose $v_1 + \cdots + v_k = 0$. Applying T^j gives

$$\lambda_1^j v_1 + \lambda_2^j v_2 + \cdots + \lambda_k^j v_k = 0 \qquad (j = 0, 1, \ldots, k-1).$$

Put the vectors in a row $[v_1 \ \cdots \ v_k]$ and the scalars in the $k \times k$ matrix M with entries $M_{ij} = \lambda_i^j$ (rows $i = 1, \ldots, k$, columns $j = 0, \ldots, k-1$). The k equations say $[v_1 \ \cdots \ v_k]\, M = [0 \ \cdots \ 0]$. This M is a Vandermonde matrix, with $\det M = \prod_{i<i'} (\lambda_{i'} - \lambda_i) \neq 0$ (*Ex*). Multiplying on the right by M^{-1} gives $[v_1 \ \cdots \ v_k] = 0$.

Theorem 1.2 by Lagrange polynomials. For each i let

$$q_i(x) = \prod_{j \neq i} \frac{x - \lambda_j}{\lambda_i - \lambda_j},$$

the polynomial of degree $k-1$ with $q_i(\lambda_i) = 1$ and $q_i(\lambda_j) = 0$ for $j \neq i$. For a λ_j-eigenvector, $q(T)v_j = q(\lambda_j)v_j$ for any polynomial q, so $q_i(T)v_j = \delta_{ij} v_j$. Applying $q_i(T)$ to $\sum_j v_j = 0$ leaves $v_i = 0$. On $\bigoplus V_\lambda$, the operator $q_i(T)$ is the projection onto V_{λ_i} along the other eigenspaces.

Proposition 1.4 from the generalized eigenspaces. After the generalized eigenspace decomposition (chapter 4), $V_\lambda = \ker(T - \lambda I) \subseteq V_\lambda^{\text{gen}}$, and $\dim V_\lambda^{\text{gen}} = m_\lambda$.

Exercises

From the board and the tutorials

1.4.1 *Sep 30 board* **[2]** Write a formal proof that eigenvectors for distinct eigenvalues are linearly independent, and that T is diagonalizable if and only if the dimensions of its eigenspaces add up to $\dim V$.

1.4.2 *Tutorial 6 Q1(a),(b)* **[1]** Decide whether each matrix is diagonalizable, over $\mathbb{R}$ and over $\mathbb{C}$; if it is, find P with $P^{-1}AP$ diagonal. $\begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{bmatrix}$; $\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$; $\begin{bmatrix} c & s \\ -s & c \end{bmatrix}$ with $c = \cos\theta$, $s = \sin\theta$, $\theta \neq 0, \pi$; $\begin{bmatrix} -2 & 0 & 1 \\ -5 & 3 & 1 \\ 4 & -2 & -1 \end{bmatrix}$; $\begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix}$; $\begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$.

1.4.3 *Tutorial 6 Q1(c)* **[1]** For a 2×2 matrix M over $\mathbb{C}$, give a necessary and sufficient condition, in terms of the entries of M, for M to be diagonalizable. Does the answer change over $\mathbb{R}$? Over $\mathbb{Q}$?

1.4.4 *Tutorial 6 Q5(a)* **[1]** Let V have basis u, v, w and let $T(u) = v$, $T(v) = w$, $T(w) = u$. Write the matrix of T. For $\mathbb{F} = \mathbb{R}$, $\mathbb{C}$ and $\mathbb{F}_p$, decide whether T is diagonalizable; over $\mathbb{F}_p$ the answer is a condition on p.

1.4.5 *Tutorial 6 Q5(b)* **[2]** A 3×3 complex matrix B has $\operatorname{tr} B = \operatorname{tr} B^2 = 0$ and $\operatorname{tr} B^3 = 3d$ with $d \neq 0$. Find $\det B$, and show that all such matrices are similar to each other.

1.4.6 *Tutorial 6 Q10, first part* **[1]** Prove that geometric multiplicity is at most algebraic multiplicity, including the fact about determinants of block triangular matrices.

1.4.7 *Oblique reflection* **[3]** Redo the oblique reflection example from earlier in the course using eigenvectors.

From the books

1.4.8 *Artin 4.4.2* **[2]** (a) Let $T^2 = I$ on a finite-dimensional V. Show that for every v, the vector $v - Tv$ is either 0 or an eigenvector with eigenvalue -1, and deduce that V is the direct sum of the eigenspaces for 1 and -1 (assume $2 \neq 0$ in $\mathbb{F}$). (b) Generalize: if $T^4 = I$ on a complex space, then V is the sum of four eigenspaces.

1.4.9 *Artin 4.4.5* **[3]** Find all invariant subspaces of the real operators with matrices $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\operatorname{diag}(1, 2, 3)$.

1.4.10 *Artin 4.4.7* **[3]** Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be real. (a) Writing the condition "X is an eigenvector" as equality of the slopes x_2/x_1 of X and of AX, find the equation satisfied by the slope s of an eigenvector. (b) If all entries of A are positive, show that A has an eigenvector in the first quadrant and one in the second quadrant.

1.4.11 *Artin 4.5.2, 4.5.3* **[2]** (a) The characteristic polynomial of $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 0 \\ 1 & * & * \end{bmatrix}$ is $t^3 - 4t - 1$. Find the missing entries. (b) Which complex numbers can be eigenvalues of an operator T with $T^r = I$? With $T^2 - 5T + 6I = 0$?

1.4.12 *Artin 4.5.5, 4.5.7, 4.5.8* **[3]** (a) Which real 2×2 matrices have real eigenvalues? Show that the eigenvalues are real when the off-diagonal entries have the same sign. (b) Do A and A^T have the same eigenvalues? The same eigenvectors? (c) For a 3×3 matrix A, show that the coefficient of t in $p_A(t)$ is the sum of the three principal 2×2 minors of A.

1.4.13 *Artin 4.6.2, 4.6.3* **[2]** (a) A complex $n \times n$ matrix A has distinct eigenvalues with eigenvectors $v_1, \ldots, v_n$. Show that every eigenvector is a multiple of some v_i, and explain how to recover A from the eigenvalues and the v_i. (b) If T has two independent eigenvectors with the same eigenvalue λ, show that λ is a multiple root of p_T.

1.4.14 *Artin 4.6.6, 4.6.7, 4.6.9* **[2]** (a) If A is diagonalizable, can the diagonalizing matrix P be chosen with $\det P = 1$? (b) If A is invertible, show that AB is similar to BA. (c) Find all real 2×2 matrices with $A^2 = I$, and describe geometrically how each acts on $\mathbb{R}^2$.

1.4.15 *Artin 4.6.11* **[3]** Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ have eigenvalue λ. (a) Show that $(b, \lambda - a)^T$, if nonzero, is an eigenvector. (b) If $b \neq 0$ and the eigenvalues are distinct, write down P with $P^{-1}AP$ diagonal.

1.4.16 *Axler 5A.23, Artin 4.M.10* **[2]** (a) Let S, T be operators on a finite-dimensional V. Prove that ST and TS have the same eigenvalues. (b) Let A be $m \times n$ and B be $n \times m$. Show that every nonzero eigenvalue of AB is an eigenvalue of BA, and that this can fail for the eigenvalue 0. Show that $I_m - AB$ is invertible if and only if $I_n - BA$ is.

1.4.17 *Axler 5A.24* **[3]** Let A be $n \times n$. (a) If every row of A sums to 1, show that 1 is an eigenvalue. (b) If every column sums to 1, show the same.

1.4.18 *Axler 5A.25, 5A.26* **[3]** (a) If u, w and $u + w$ are all eigenvectors of T, show that u and w have the same eigenvalue. (b) If every nonzero vector is an eigenvector of T, show that T is a scalar multiple of I (Artin 4.4.8).

1.4.19 *Axler 5A.27* **[3]** Let $1 \leq k \leq \dim V - 1$. If every k-dimensional subspace of V is T-invariant, prove that T is a scalar multiple of I.

1.4.20 *Axler 5A.34* **[3]** Show that a list $v_1, \ldots, v_m$ in a finite-dimensional V is linearly independent if and only if there is an operator T on V for which $v_1, \ldots, v_m$ are eigenvectors with distinct eigenvalues.

1.4.21 *Axler 5A.35, 5A.36* **[3]** (a) For distinct reals $\lambda_1, \ldots, \lambda_n$, show that $e^{\lambda_1 x}, \ldots, e^{\lambda_n x}$ are linearly independent functions. (Hint: differentiation on their span.) (b) For distinct positive $\lambda_1, \ldots, \lambda_n$, show that $\cos(\lambda_1 x), \ldots, \cos(\lambda_n x)$ are linearly independent.

1.4.22 *Axler 5D.6, 5D.9, 5D.10* **[1]** (a) Let T be an operator on $\mathbb{F}^5$ whose 8-eigenspace has dimension 4. Prove that $T - 2I$ or $T - 6I$ is invertible. (b) If R, T are operators on $\mathbb{F}^3$ that both have eigenvalues $2, 6, 7$, show that $R = S^{-1}TS$ for some invertible S. (c) Find R, T on $\mathbb{F}^4$, each with exactly the eigenvalues $2, 6, 7$, that are not similar.

1.4.23 *Axler 5D.13* **[2]** If A is diagonal with distinct diagonal entries, show that $AB = BA$ if and only if B is diagonal.

1.4.24 *Axler 5D.21* **[1]** Let $T(x, y) = (y, x + y)$ on $\mathbb{R}^2$ and let F_n be the Fibonacci numbers ($F_0 = 0$, $F_1 = 1$). Show $T^n(0, 1) = (F_n, F_{n+1})$, diagonalize T, derive Binet's formula, and show that F_n is the integer nearest to $\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n$.

Harder or unusual

1.4.25 *Napkin 10E* **[4]** Let V be the complex polynomials of degree at most 2, and $T(ax^2 + bx + c) = cx^2 + bx + a$. Find the eigenspaces of T.

1.4.26 *Axler 5B.4, 5B.5* **[3]** Let $\mathbb{F} = \mathbb{C}$ and p a nonconstant polynomial. Prove that α is an eigenvalue of $p(T)$ if and only if $\alpha = p(\lambda)$ for some eigenvalue λ of T. Give an operator on $\mathbb{R}^2$ showing that this fails over $\mathbb{R}$.

1.4.27 *Artin 5.M.7* **[3]** Let A be a 2×2 complex matrix with distinct eigenvalues. How many solutions X does $X^2 = A$ have?

1.4.28 *Napkin 12F, Putnam 2015* **[4]** Let S be the set of real matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with a, b, c, d in arithmetic progression in that order. Find all $M \in S$ such that $M^k \in S$ for some integer $k > 1$.

1.4.29 *Axler 5A.37* **[2]** Let $T \in L(V)$ and define $\mathcal{A}$ on $L(V)$ by $\mathcal{A}(S) = TS$. Prove that T and $\mathcal{A}$ have the same eigenvalues.

1.4.30 **[4]** Let A be a real $n\times n$ matrix with $A^2 = -I$. Show that n is even, and that A is similar over $\mathbb{R}$ to the block diagonal matrix with $n/2$ blocks $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

Triangular Form and Nilpotent Operators

2

2.1 Invariant subspaces and restrictions

Definition 2.1

A subspace W of V is *T-stable* (or *T-invariant*) if $T(W) \subseteq W$, that is, $Tw \in W$ for every $w \in W$.

Definition 2.2

If W is T-invariant, the *restriction* of T to W is the map

$$T|_W : W \to W, \qquad T|_W(w) = Tw.$$

It does the same thing as T, but only to vectors of W. It is a linear operator on W *because* W is invariant: without invariance, Tw could leave W, and $T|_W$ would not be a map from W to W.

Example 2.1

Let $T(x, y, z) = (2x + y,\ 3y,\ 5z)$ on $\mathbb{R}^3$ and $W = \operatorname{span}(e_1, e_2)$, the xy-plane. Then $T(x, y, 0) = (2x + y, 3y, 0) \in W$, so W is invariant, and in the basis e_1, e_2 of W the restriction has matrix $\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$. On the other hand $W' = \operatorname{span}(e_2)$ is not invariant, since $Te_2 = (1, 3, 0) \notin W'$.

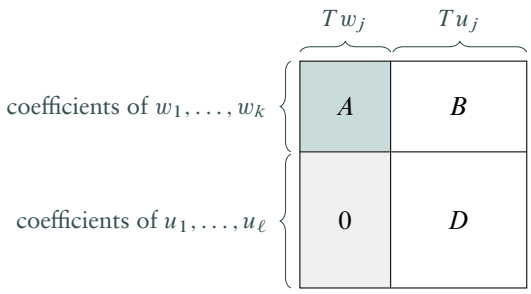
Lemma 2.1

Block form

Let W be T-invariant with basis $w_1, \dots, w_k$, extended by $u_1, \dots, u_\ell$ to a basis of V. In the basis $w_1, \dots, w_k, u_1, \dots, u_\ell$,

$$[T] = \begin{bmatrix} A & B \\ 0 & D \end{bmatrix}, \qquad \text{where } A = [T|_W] \text{ is } k \times k \text{ and } D \text{ is } \ell \times \ell.$$

Consequently $p_T(x) = p_A(x)\, p_D(x) = p_{T|_W}(x)\, p_D(x)$.



The first k columns are the vectors Tw_j, which lie in W, so they have no u-coefficients: that is the zero block.

Proof. The j-th column of $[T]$ holds the coordinates of Tw_j. Since $Tw_j \in W = \operatorname{span}(w_1, \dots, w_k)$, its coordinates along $u_1, \dots, u_\ell$ are 0; this is the zero block. The top k entries of that column are the coordinates of $T|_W(w_j)$ in the basis $w_1, \dots, w_k$, so the top-left block is $[T|_W]$. Finally $xI - [T] = \begin{bmatrix} xI_k - A & -B \\ 0 & xI_\ell - D \end{bmatrix}$ is block upper triangular, so its determinant is $\det(xI_k - A)\det(xI_\ell - D)$ by proposition 3.6. □

If the basis can be chosen so that $B = 0$ as well, then $U = \operatorname{span}(u_1, \dots, u_\ell)$ is T-stable too, and T breaks into the two separate operators $T|_W$ and $T|_U$. The best case in this direction is an eigenbasis: a 1-dimensional T-stable subspace is the same thing as the line through an eigenvector.

Remark 2.1.1.

1. For every scalar λ, a subspace is T-stable if and only if it is $(T - \lambda I)$-stable, since $(T - \lambda I)w = Tw - \lambda w$ and $\lambda w \in W$.
2. If $ST = TS$, then $\ker S$ and $\operatorname{im} S$ are T-stable: if $Sv = 0$ then $S(Tv) = T(Sv) = 0$, and $T(Sv) = S(Tv) \in \operatorname{im} S$. In particular $\ker p(T)$ and $\operatorname{im} p(T)$ are T-stable for every polynomial p, since $p(T)$ commutes with T.

2.2 Triangular form

Recall that p_T *factors fully* over $\mathbb{F}$ if $p_T(x) = (x - \lambda_1)(x - \lambda_2)\cdots(x - \lambda_n)$ for some $\lambda_i \in \mathbb{F}$, not necessarily distinct. Over $\mathbb{C}$ this always holds; over $\mathbb{R}$ it can fail, as for rotation by $\pi/2$.

Triangular form **Theorem 2.2**

T has an upper triangular matrix in some basis if and only if p_T factors fully over $\mathbb{F}$. In that case, the diagonal entries are the roots of p_T, each repeated according to its multiplicity.

Proof of "triangular ⇒ factors fully". If $[T]$ is upper triangular with diagonal entries $a_{11}, \dots, a_{nn}$, then $xI - [T]$ is upper triangular with diagonal entries $x - a_{ii}$, and the determinant of a triangular matrix is the product of its diagonal entries. So $p_T(x) = \prod_i (x - a_{ii})$, which factors fully, with roots the a_{ii}. □

Proof of "factors fully ⇒ triangular" (class). In matrix form, the claim is:

> $(*)_n$: every $n \times n$ matrix A whose characteristic polynomial factors fully is similar to an upper triangular matrix, that is, $P^{-1}AP$ is upper triangular for some invertible P.

We prove $(*)_n$ by induction on n. For $n = 1$ every matrix is upper triangular. Assume $(*)_{n-1}$ and let A be $n \times n$ with $p_A(x) = (x - \lambda_1) \cdots (x - \lambda_n)$.

Step 1: put an eigenvector first. Since $p_A(\lambda_1) = 0$, λ_1 is an eigenvalue; let v_1 be a λ_1-eigenvector and extend it to a basis $v_1, v_2, \dots, v_n$ of $\mathbb{F}^n$. Let $P_1 = [v_1 \cdots v_n]$. The first column of $P_1^{-1}AP_1$ holds the coordinates of $Av_1 = \lambda_1 v_1$, namely $(\lambda_1, 0, \dots, 0)^T$. So

$$P_1^{-1}AP_1 = \begin{bmatrix} \lambda_1 & r \\ 0 & B \end{bmatrix},$$

where r is some row of length $n - 1$ and B is an $(n-1) \times (n-1)$ matrix. (This is lemma 2.1 for the invariant line $W = \mathbb{F}v_1$.)

Step 2: the smaller matrix B also factors fully. Similar matrices have the same characteristic polynomial, and by lemma 2.1

$$(x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n) = p_A(x) = (x - \lambda_1)\, p_B(x).$$

Cancel $x - \lambda_1$ (polynomials can be cancelled) to get $p_B(x) = (x - \lambda_2) \cdots (x - \lambda_n)$.

Step 3: apply induction to B. By $(*)_{n-1}$ there is an invertible $(n-1) \times (n-1)$ matrix Q with $Q^{-1}BQ$ upper triangular.

Step 4: combine. Let $P_2 = \begin{bmatrix} 1 & 0 \\ 0 & Q \end{bmatrix}$, which is invertible with $P_2^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & Q^{-1} \end{bmatrix}$. Multiplying blocks,

$$P_2^{-1} \begin{bmatrix} \lambda_1 & r \\ 0 & B \end{bmatrix} P_2 = \begin{bmatrix} 1 & 0 \\ 0 & Q^{-1} \end{bmatrix} \begin{bmatrix} \lambda_1 & rQ \\ 0 & BQ \end{bmatrix} = \begin{bmatrix} \lambda_1 & rQ \\ 0 & Q^{-1}BQ \end{bmatrix},$$

which is upper triangular because $Q^{-1}BQ$ is. So $P = P_1P_2$ works. The diagonal entries are λ_1 followed by those of $Q^{-1}BQ$, which by induction are $\lambda_2, \dots, \lambda_n$. □

2.3 Nilpotent operators

Definition 2.3

T is *nilpotent* if $T^N = 0$ for some $N > 0$. A matrix is *strictly upper triangular* if it is upper triangular with zeros on the diagonal.

Example 2.2

Let J be the $n \times n$ matrix with 1s just above the diagonal and 0 elsewhere, so that $Je_1 = 0$ and $Je_i = e_{i-1}$: the chain $e_n \mapsto e_{n-1} \mapsto \cdots \mapsto e_1 \mapsto 0$. Then $J^n = 0$ but $J^{n-1} \neq 0$, since $J^{n-1}e_n = e_1$.

Lemma 2.3

Kernel chain

For any operator T, $0 = \ker T^0 \subseteq \ker T \subseteq \ker T^2 \subseteq \cdots$. If $\ker T^k = \ker T^{k+1}$ for some k, then $\ker T^k = \ker T^{k+j}$ for every $j \geq 0$. Hence the dimensions $\dim \ker T^i$ strictly increase until they become constant, and they become constant by $i = n$.

Proof. If $T^i v = 0$ then $T^{i+1}v = T(T^i v) = 0$, so the kernels increase. Now suppose $\ker T^k = \ker T^{k+1}$, and let $v \in \ker T^{k+2}$. Then $T^{k+1}(Tv) = 0$, so $Tv \in \ker T^{k+1} = \ker T^k$, so $T^{k+1}v = T^k(Tv) = 0$, so $v \in \ker T^{k+1}$. Thus $\ker T^{k+2} = \ker T^{k+1}$, and induction on j finishes. Finally, while the chain is strictly increasing, each step raises the dimension by at least 1; the dimension cannot exceed n, so there are at most n strict steps. □

Theorem 2.4

Oct 6–7

(A) If T is nilpotent and diagonalizable, then $T = 0$.

(B) The following are equivalent:

(1) T is nilpotent;

(2) T has a strictly upper triangular matrix in some basis;

(3) $p_T(x) = x^n$;

(4) $T^n = 0$.

(C) (= theorem 2.2) p_T factors fully $\iff$ T has a triangular matrix.

Proof of (A). Let $T^N = 0$ and let $Tv = \lambda v$ with $v \neq 0$. Applying T repeatedly gives $T^k v = \lambda^k v$ for every k, so $0 = T^N v = \lambda^N v$. Since $v \neq 0$, $\lambda^N = 0$, so $\lambda = 0$. Thus 0 is the only eigenvalue. If T is also diagonalizable, its diagonal matrix has the eigenvalues on the diagonal, so it is the zero matrix, and $T = 0$. □

Proof of (B). We prove (1) ⇒ (4) ⇒ (1), (1) ⇒ (2) ⇒ (3) ⇒ (2), and (2) ⇒ (4).

(4) ⇒ (1). Take $N = n$.

(1) ⇒ (4). If $T^N = 0$, then $\ker T^N = V$, so the kernel chain of lemma 2.3 reaches V. By the lemma it stops growing by step n, so $\ker T^n = \ker T^N = V$, that is, $T^n = 0$.

(1) ⇒ (2) *(class).* Write $K_i = \ker T^i$. By lemma 2.3, $0 = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n = V$. Choose a basis of K_1, extend it to a basis of K_2, then to a basis of K_3, and so on up to V; call the result $v_1, \ldots, v_n$, listed in the order chosen. The key fact is

$$T(K_i) \subseteq K_{i-1} \quad (i \geq 1),$$

because if $T^i v = 0$ then $T^{i-1}(Tv) = 0$. Now take a basis vector v_j, and let i be the first index with $v_j \in K_i$. Then $Tv_j \in K_{i-1}$, and K_{i-1} is spanned by basis vectors chosen *before* v_j. So Tv_j is a combination of $v_1, \ldots, v_{j-1}$ only: the j-th column of $[T]$ has zeros on and below the diagonal. So $[T]$ is strictly upper triangular.

(2) ⇒ (3). By the first half of the proof of theorem 2.2, $p_T(x) = \prod_i (x - 0) = x^n$.

(3) ⇒ (2). $p_T = x^n$ factors fully with every root 0, so by theorem 2.2 T has an upper triangular matrix with zeros on the diagonal.

(2) ⇒ (4). Let $[T]$ be strictly upper triangular in the basis $v_1, \ldots, v_n$ and $V_k = \operatorname{span}(v_1, \ldots, v_k)$, $V_0 = 0$. Column k says $Tv_k \in V_{k-1}$, so $T(V_k) \subseteq V_{k-1}$ for each k. Applying T n times: $T^n(V) = T^n(V_n) \subseteq V_0 = 0$. □

2.4 Other proofs

Theorem 2.2 by invariant hyperplanes. This builds the triangular form from the top instead of from the bottom. Let A be $n \times n$ with p_A factoring fully.

1. *A^T has the same characteristic polynomial.* Since $\det M^T = \det M$ for every square matrix and $(xI - A)^T = xI - A^T$, we get $p_{A^T}(x) = \det(xI - A^T) = \det\left((xI - A)^T\right) = \det(xI - A) = p_A(x)$. So λ_n is an eigenvalue of A^T; let $w \neq 0$ with $A^T w = \lambda_n w$.

2. *A hyperplane that A preserves.* Let $H = \{x \in \mathbb{F}^n : w^T x = 0\}$, a subspace of dimension $n - 1$ (one nonzero linear equation). For $x \in H$, $w^T(Ax) = (A^T w)^T x = \lambda_n\, w^T x = 0$, so $Ax \in H$. Thus H is A-invariant, and the restriction $A|_H \colon H \to H$, $x \mapsto Ax$, is an operator on H.

3. *Its characteristic polynomial.* Take a basis $h_1, \ldots, h_{n-1}$ of H and a vector $u \notin H$. In the basis $h_1, \ldots, h_{n-1}, u$, lemma 2.1 gives $\begin{bmatrix} [A|_H] & * \\ 0 & c \end{bmatrix}$ for a scalar c, meaning $Au = h + cu$ with $h \in H$. Apply w^T: since $w^T h = 0$, we get $w^T Au = c\, w^T u$; also $w^T Au = (A^T w)^T u = \lambda_n w^T u$. As $u \notin H$, $w^T u \neq 0$, so $c = \lambda_n$. So $p_A(x) = p_{A|_H}(x)(x - \lambda_n)$, and $p_{A|_H}$ factors fully.

4. *Induction.* By induction there is a basis $h_1, \dots, h_{n-1}$ of H in which $A|_H$ is upper triangular. Add $u \notin H$ at the end. Columns $1, \dots, n-1$ are those of $[A|_H]$ padded with a 0, and column n can be anything, so the matrix is upper triangular.

Theorem 2.2 as a statement about flags. A basis $v_1, \dots, v_n$ makes $[T]$ upper triangular if and only if each $V_k = \operatorname{span}(v_1, \dots, v_k)$ is T-invariant, if and only if $Tv_k \in V_k$ for each k (column k has zeros below row k; Axler 5C.10). Since the v_k are independent, $\dim V_k = k$. A chain $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$ of subspaces with $\dim V_k = k$ is called a *full flag*; conversely, choosing $v_k \in V_k \setminus V_{k-1}$ for each k turns a full flag into a basis. So theorem 2.2 says: a full flag of T-invariant subspaces exists if and only if p_T factors fully. Neither proof mentions flags, but each produces one, from the triangularizing basis it constructs. The class proof fixes $V_1 = \mathbb{F}v_1$ first and gets the rest by induction; the hyperplane proof fixes $V_{n-1} = H$ first and gets the rest by induction inside H.

(B), (1) $\Rightarrow$ (2) *by images.* Let $T^N = 0$ and use the decreasing chain $V \supseteq T(V) \supseteq T^2(V) \supseteq \cdots \supseteq T^N(V) = 0$. Choose a basis of the smallest nonzero term, extend it to a basis of the next larger term, and continue up to V; list the vectors in the order chosen. If v_j was chosen for $T^i(V)$ but not earlier, then $Tv_j \in T^{i+1}(V)$, which is spanned by vectors chosen before v_j. So again column j has zeros on and below the diagonal (*Ex*).

(B), (1) $\Rightarrow$ (4) *by Cayley–Hamilton.* If T is nilpotent, then by (A) its only possible eigenvalue is 0, also over $\mathbb{C}$ for a real or complex matrix. Over $\mathbb{C}$, p_T factors fully, so all its roots are eigenvalues, hence all are 0, so $p_T = x^n$; Cayley–Hamilton (section 3.3) gives $T^n = p_T(T) = 0$. This uses more machinery than lemma 2.3, and over a general field one needs the roots in a larger field, so the chain argument is the better proof.

Exercises

From the board and the tutorials

2.4.1 *Oct 7 board* **[2]** Show that $\ker T^k = \ker T^{k+1}$ implies $\ker T^k = \ker T^{k+j}$ for all $j \geq 0$, and that the same holds for images. Write a precise proof by following your nose.

2.4.2 *Sep 30 board* **[2]** Show that no nonzero nilpotent matrix is diagonalizable.

2.4.3 *Oct 6 board* **[2]** Complete the proof of (B), including the version with the decreasing chain of images.

From the books

2.4.4 *Artin 4.4.6, 4.6.8* **[1]** (a) Let P be the real polynomials of degree at most n and $D = \frac{d}{dx}$. Show that D is nilpotent, find its matrix in a convenient basis, and find all D-invariant subspaces. (b) Prove that T is nilpotent if and only if some basis makes its matrix upper triangular with zero diagonal.

2.4.5 *Axler 5C.1, 5C.3* **[3]** (a) Prove or give a counterexample: if T^2 has an upper triangular matrix in some basis, then so does T. (b) If T is invertible and upper triangular in the basis $v_1, \dots, v_n$ with diagonal $\lambda_1, \dots, \lambda_n$, show that T^{-1} is upper triangular in the same basis with diagonal $1/\lambda_1, \dots, 1/\lambda_n$.

2.4.6 *Axler 5C.4, 5C.5* **[3]** Give an invertible operator whose matrix in some basis has only 0s on the diagonal, and a non-invertible operator whose matrix in some basis has only nonzero entries on the diagonal. (So "invertible iff no zero on the diagonal" needs the matrix to be triangular.)

2.4.7 *Axler 5C.6* **[2]** Let $\mathbb{F} = \mathbb{C}$. Prove that for every $k \in \{1, \dots, \dim V\}$, V has a k-dimensional T-invariant subspace.

2.4.8 *Axler 5C.10, 5C.11* **[3]** Show that the matrix of T in the basis $v_1, \dots, v_n$ is lower triangular if and only if each $\operatorname{span}(v_k, \dots, v_n)$ is T-invariant. Deduce that over $\mathbb{C}$ every operator has a lower triangular matrix in some basis.

2.4.9 *Axler 8A.1, 8A.2* **[1]** (a) If $\dim\ker T^4 = 8$ and $\dim\ker T^6 = 9$, prove that $\dim\ker T^m = 9$ for all $m \geq 5$. (b) If $T^{m-1}v \neq 0$ but $T^m v = 0$, prove that $v, Tv, \dots, T^{m-1}v$ are linearly independent.

2.4.10 *Axler 8A.3, 8A.5* **[2]** (a) Prove that $V = \ker T \oplus \operatorname{im} T$ if and only if $\ker T^2 = \ker T$. (b) Prove that $\dim\ker T^m \leq m \dim\ker T$.

2.4.11 *Axler 8A.13, 8A.16* **[3]** (a) If ST is nilpotent, prove that TS is nilpotent. (b) Give nilpotent S, T such that neither $S + T$ nor ST is nilpotent. If S, T are nilpotent and $ST = TS$, prove that $S + T$ and ST are nilpotent.

2.4.12 *Axler 8A.17, 8A.18* **[3]** Let T be nilpotent with $T^m = 0$. (a) Prove that $I - T$ is invertible with inverse $I + T + \cdots + T^{m-1}$. How would you guess this formula? (b) Prove that $T^{1+\dim\operatorname{im} T} = 0$.

2.4.13 *Axler 8A.21, 8A.22, 8A.23* **[2]** (a) If $\ker T^{n-1} \neq \ker T^n$ where $n = \dim V$, prove that T is nilpotent and $\dim\ker T^k = k$ for $0 \leq k \leq n$. (b) If T is an operator on $\mathbb{C}^5$ with $\operatorname{im} T^4 \neq \operatorname{im} T^5$, prove that T is nilpotent. (c) Give an operator on a real vector space whose only eigenvalue is 0 but which is not nilpotent.

2.4.14 *Halmos §57.1, §57.2* **[2]** (a) Can a nilpotent operator on a 2-dimensional space satisfy $T^2 \neq 0$? (b) Show that a nilpotent operator has trace 0. (c) Over $\mathbb{R}$ or $\mathbb{C}$, let $C = AB - BA$. Show that $I - C$ is not nilpotent. In particular $AB - BA = I$ is impossible. What goes wrong over $\mathbb{F}_p$?

Harder or unusual

2.4.15 *Halmos §56.2* **[3]** Over $\mathbb{C}$, if $AB = BA$, prove that there is one basis in which both A and B are upper triangular. (Hint: an eigenspace of A is B-invariant, so it contains a common eigenvector.)

2.4.16 *Axler 5E.9* **[4]** Let $\mathscr{C}$ be any set (possibly infinite) of pairwise commuting operators on a nonzero finite-dimensional complex space. Prove that some vector is an eigenvector of every element of $\mathscr{C}$, and that some basis makes every element of $\mathscr{C}$ upper triangular.

2.4.17 *Artin 5.M.3* **[4]** Let A be a complex $n \times n$ matrix with $\operatorname{tr}(A^k) = 0$ for every $k \geq 1$. Prove that A is nilpotent. (Hint: triangularize, and use the power sums of the eigenvalues.)

2.4.18 *Axler 8A.25* **[4]** If V is an inner product space and T is nilpotent, show that some orthonormal basis makes the matrix of T upper triangular with zero diagonal. (Use corollary 5.7.)

Cayley–Hamilton, Minimal Polynomials, Tools

3

These results are not all on the boards, but Tutorial 6 and the 2022 HW6 use them, and the instructor put all tutorial problems (except those on excluded proofs) in the syllabus. For a polynomial $q(x) = \sum c_k x^k$, we write $q(T) = \sum c_k T^k$ (with $T^0 = I$). Two facts are used throughout: $q(T)$ commutes with T, and if $Tv = \lambda v$ then $q(T)v = q(\lambda)v$.

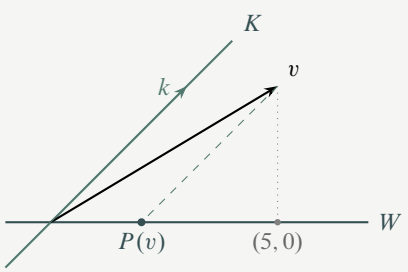
3.1 Projections

Definition 3.1

An operator P on V is a *projection* if $P^2 = P$. (This is not yet an *orthogonal* projection, which needs an inner product.) Given $V = K \oplus W$, every $v \in V$ is $v = k + w$ for *unique* $k \in K$, $w \in W$, and the *projection onto W along K* is $P(v) = w$, the W-component of v. Equivalently, P is the linear map with $P(k) = 0$ for $k \in K$ and $P(w) = w$ for $w \in W$. It depends on K as well as on W.

Example 3.1

In $\mathbb{R}^2$ let W be the x-axis and $K = \mathbb{R}(1,1)^T$. Then $v = (5,3)^T$ splits as $v = (3,3)^T + (2,0)^T$ with $(3,3)^T \in K$ and $(2,0)^T \in W$, so $P(v) = (2,0)^T$: we slide v to W *parallel to* K. Sliding perpendicular to W instead would give $(5,0)^T$, the orthogonal projection, which is the projection along $K' = W^\perp$, the y-axis.



Proposition 3.1 (Tutorial 6 Q4).

1. $P^2 = P \iff P$ *is the identity on* $\operatorname{im} P$.
2. *If* $P^2 = P$, *then* $V = \ker P \oplus \operatorname{im} P$, P *is diagonalizable with eigenvalues in* $\{0, 1\}$, *and* $\operatorname{tr} P = \operatorname{rank} P$ *(as an element of* $\mathbb{F}$*)*.
3. $V = K \oplus W$ *if and only if there is a projection* P *with* $\ker P = K$ *and* $\operatorname{im} P = W$.
4. *Let* K *be* T*-invariant. There is a* T*-invariant* W *with* $V = K \oplus W$ *if and only if there is a projection* P *with* $\ker P = K$ *and* $TP = PT$.

Proof. (a) $P^2 = P$ says $P(Pv) = Pv$ for all v, that is, P fixes every vector of the form Pv.

(b) Every v splits as $v = (v - Pv) + Pv$, with $P(v - Pv) = Pv - P^2v = 0$, so $V = \ker P + \operatorname{im} P$. If $v \in \ker P \cap \operatorname{im} P$, then $v = Pv$ by (a) and $Pv = 0$, so $v = 0$. A basis of $\ker P$ followed by a basis of $\operatorname{im} P$ is an eigenbasis with eigenvalues 0 and 1; the diagonal matrix has rank P ones, so $\operatorname{tr} P = \operatorname{rank} P$.

(c) Given $V = K \oplus W$, the projection onto W along K works; conversely use (b).

(d) If $TP = PT$, then $\ker P$ and $\operatorname{im} P$ are T-invariant (Remark in §2.1), so $W = \operatorname{im} P$ works. Conversely, if $V = K \oplus W$ with both invariant and P is the projection onto W along K, check $TP(k + w) = Tw = PT(k + w)$, using $Tk \in K$ and $Tw \in W$ (*Ex*). □

3.2 Invariant subspaces of diagonalizable operators

Tutorial 6 Q7 — **Theorem 3.2**

1. Let T be diagonalizable, so $V = \bigoplus_\lambda V_\lambda$. Every T-invariant subspace U satisfies $U = \bigoplus_\lambda (U \cap V_\lambda)$. In particular $T|_U$ is diagonalizable.
2. If S, T are diagonalizable and $ST = TS$, there is one basis in which both are diagonal. The same holds for any family of mutually commuting diagonalizable operators.

Proof. (a) The inclusion $\supseteq$ is clear. For $\subseteq$, let $u \in U$ and write $u = \sum_\lambda u_\lambda$ with $u_\lambda \in V_\lambda$. We must show each $u_\lambda \in U$. Fix λ and apply $q(T) = \prod_{\mu \neq \lambda} (T - \mu I)$, the product over the other eigenvalues. It kills each u_μ with $\mu \neq \lambda$ and multiplies u_λ by the nonzero scalar $c = \prod_{\mu \neq \lambda} (\lambda - \mu)$. So $q(T)u = cu_\lambda$. Since U is invariant, $q(T)u \in U$, so $u_\lambda = c^{-1} q(T)u \in U$. Bases of the $U \cap V_\lambda$ together form an eigenbasis of U.

(b) Each eigenspace V_λ of T is S-invariant: if $Tv = \lambda v$ then $T(Sv) = S(Tv) = \lambda Sv$. So $S|_{V_\lambda}\colon V_\lambda \to V_\lambda$ is an operator on V_λ, and by (a) applied to the diagonalizable S and its invariant subspace V_λ, it is diagonalizable: V_λ has a basis of eigenvectors of $S|_{V_\lambda}$, that is, of vectors $u \in V_\lambda$ with $Su = \mu u$ for some μ. Choose such a basis in each V_λ. Together these vectors are a basis of V, they are eigenvectors of S, and they are eigenvectors of T since they lie in eigenspaces of T.

A family. Let $\mathcal{F}$ be any set (possibly infinite) of pairwise commuting diagonalizable operators on V. We show by induction on $\dim V$ that some basis of V consists of common eigenvectors of all members of $\mathcal{F}$. If every member of $\mathcal{F}$ is a scalar multiple of I, every vector is an eigenvector of every member, and any basis works. Otherwise pick a member T that is not a scalar, and split $V = \bigoplus_\lambda V_\lambda$ into its eigenspaces. Since T is not a scalar, it has at least two distinct eigenvalues, so each V_λ has dimension $< \dim V$. Each V_λ is invariant under every $S \in \mathcal{F}$, by the argument above. The restrictions $S|_{V_\lambda}$ ($S \in \mathcal{F}$) are diagonalizable by (a) and still commute, so by induction V_λ has a basis of common eigenvectors of all of them. Putting these bases together gives a basis of V of common eigenvectors. (For a finite family one can also just repeat the two-operator argument.) The converse holds: diagonal matrices commute. □

3.3 The Cayley–Hamilton theorem

Cayley–Hamilton **Theorem 3.3**

Every operator satisfies its characteristic polynomial: $p_T(T) = 0$.

Proof 1 (Tutorial 6 Q8): triangular form. Diagonalizable T. For an eigenvector, $p_T(T)v = p_T(\lambda)v = 0$, since λ is a root of p_T. So $p_T(T)$ kills an eigenbasis, hence is 0.

Triangularizable T. Let the matrix be upper triangular with diagonal $\lambda_1, \dots, \lambda_n$ in the basis $v_1, \dots, v_n$, and let $V_k = \operatorname{span}(v_1, \dots, v_k)$, $V_0 = 0$. Column k of the matrix says $Tv_k = \lambda_k v_k +$ (combination of $v_1, \dots, v_{k-1}$), so

$$(T - \lambda_k I)V_k \subseteq V_{k-1}.$$

Since $p_T(T) = (T - \lambda_1 I)(T - \lambda_2 I)\cdots(T - \lambda_n I)$ and these factors commute, apply them to $V = V_n$ in the order $T - \lambda_n I$ first: $V_n \to V_{n-1} \to \cdots \to V_0 = 0$.

Over $\mathbb{C}$ every operator is triangularizable (theorem 2.2), so this proves the theorem for every real or complex matrix. □

Proof 2 (Tutorial 6 Q9): cyclic subspaces. This works over any field. The idea is to build a basis in stages, each stage being a *cyclic* piece $v, Tv, T^2v, \dots$.

The companion matrix. Let $v_0 \neq 0$. Apply T repeatedly until $T^{m+1}v_0$ first depends on $v_0, Tv_0, \dots, T^m v_0$, say

$$T^{m+1}v_0 = a_0 v_0 + a_1 T v_0 + \cdots + a_m T^m v_0.$$

Then $U_1 = \operatorname{span}(v_0, \dots, T^m v_0)$ is T-invariant, and in this basis the matrix of $T|_{U_1}$ has 1s just below the diagonal and last column $(a_0, \dots, a_m)^T$ (Artin 4.5.6). Its characteristic polynomial is $q_1(x) = x^{m+1} - a_m x^m - \cdots - a_0$ (*Ex*). By construction $q_1(T)v_0 = 0$, and then $q_1(T)(T^i v_0) = T^i q_1(T) v_0 = 0$, so $q_1(T)$ kills U_1.

The next stages. If $U_1 \neq V$, pick $w_0 \notin U_1$ and apply T while the new vector stays independent of *all* vectors chosen so far. This gives $U_2 \supsetneq U_1$, again T-invariant. As before, the relation that stops the process reads $q_2(T)w_0 \in U_1$ for a monic polynomial q_2, and then $q_2(T)U_2 \subseteq U_1$. Continue to $U_1 \subsetneq U_2 \subsetneq \cdots \subsetneq U_s = V$.

Conclusion. In the basis built this way, the matrix of T is block upper triangular, with the companion matrix of q_j as the j-th diagonal block (*Ex*). So $p_T = q_1 q_2 \cdots q_s$ by proposition 3.6, and

$$p_T(T)V = q_1(T)\cdots q_s(T)U_s \subseteq q_1(T)\cdots q_{s-1}(T)U_{s-1} \subseteq \cdots \subseteq q_1(T)U_1 = 0. \qquad \square$$

Proof 3: the adjugate. Let A be the matrix and $p(x) = \det(xI - A) = \sum_{k=0}^{n} c_k x^k$ (so $c_n = 1$). For any square matrix M, $M \operatorname{adj}(M) = \det(M) I$; apply this to $M = xI - A$:

$$(xI - A)\operatorname{adj}(xI - A) = p(x) I.$$

Each entry of $\operatorname{adj}(xI - A)$ is a $(n-1)\times(n-1)$ minor of $xI - A$, a polynomial of degree at most $n-1$. Collecting powers of x, $\operatorname{adj}(xI - A) = \sum_{k=0}^{n-1} B_k x^k$ for constant matrices B_k. Expanding the left side and comparing coefficients of x^k:

$$-AB_0 = c_0 I, \qquad B_{k-1} - AB_k = c_k I \ \ (1 \le k \le n-1), \qquad B_{n-1} = I.$$

Multiply the equation for x^k on the left by A^k and add: on the left the terms cancel in pairs ($A^k B_{k-1}$ against $-A^k B_{k-1}$ from the previous equation), and on the right we get $\sum c_k A^k = p(A)$. So $p(A) = 0$. This works over every field. $\square$

Proof 4: Jordan form (Artin 5.M.2). Over $\mathbb{C}$, write $A = PJP^{-1}$ with J in Jordan form. Then $p_A(A) = P\, p_A(J)\, P^{-1}$, and $p_A(J)$ is block diagonal with blocks $p_A(J_{\lambda,d})$. The factor $(J_{\lambda,d} - \lambda I)^{m_\lambda}$ of $p_A(J_{\lambda,d})$ is 0, since $J_{\lambda,d} - \lambda I$ is a shift with d-th power 0 and $d \le m_\lambda$. $\square$

Proof 5: continuity (Artin §5.2). Matrices with n distinct eigenvalues are diagonalizable, and they are dense in $M_n(\mathbb{C})$: triangularize A and change the diagonal entries slightly to make them distinct. For such matrices the theorem holds by Proof 1. Both sides of $p_A(A) = 0$ are polynomial, hence continuous, functions of the entries of A, so the identity extends to all A (*Ex*). $\square$

Remark 3.3.1. The "proof" $p_A(A) = \det(AI - A) = \det(0) = 0$ is wrong. In $p_A(x) = \det(xI - A)$ the variable x is a scalar: $\det(xI - A)$ is a number for each x, while $p_A(A)$ is a matrix. Substituting a matrix for x inside the determinant is a different operation from evaluating the polynomial p_A at A.

Example 3.2

If A is 3×3 with $p_A(x) = x^3 - 2x + 1$, then $A^3 - 2A + I = 0$, so $A(A^2 - 2I) = -I$ and $A^{-1} = 2I - A^2$.

3.4 The minimal polynomial

Definition 3.2

The *minimal polynomial* m_T is the monic polynomial of least degree with $m_T(T) = 0$. It exists: the $n^2 + 1$ operators $I, T, T^2, \dots, T^{n^2}$ lie in the n^2-dimensional space $L(V)$, so some nontrivial combination of them is 0. (By Cayley–Hamilton, even degree n suffices.)

Proposition 3.4.

1. *If $f(T) = 0$, then m_T divides f. In particular, m_T divides p_T.*
2. *m_T and p_T have the same roots, namely the eigenvalues.*
3. *T is diagonalizable if and only if m_T is a product of distinct linear factors.*
4. *If $p_T = \prod(x - \lambda_i)^{m_i}$, then $m_T = \prod(x - \lambda_i)^{s_i}$, where s_i is the size of the largest Jordan block for λ_i.*

Proof. (a) Divide with remainder: $f = q m_T + r$ with $\deg r < \deg m_T$. Then $r(T) = f(T) - q(T)m_T(T) = 0$. A nonzero r (made monic) would contradict the minimality of $\deg m_T$, so $r = 0$.

(b) If $Tv = \lambda v$ with $v \neq 0$, then $0 = m_T(T)v = m_T(\lambda)v$, so $m_T(\lambda) = 0$. Conversely, every root of m_T is a root of p_T by (a).

(c, $\Rightarrow$) Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T and $g(x) = (x-\lambda_1)\cdots(x-\lambda_k)$. Take an eigenbasis. A basis vector v with $Tv = \lambda_i v$ is killed by the factor $T - \lambda_i I$ of $g(T)$, and the factors commute, so $g(T)v = 0$. Thus $g(T)$ kills a basis, so $g(T) = 0$, and by (a) m_T divides g. A monic divisor of a product of distinct linear factors is again a product of distinct linear factors.

(c, $\Leftarrow$) Suppose $m_T(x) = (x - \lambda_1)(x - \lambda_2)\cdots(x - \lambda_k)$ with $\lambda_1, \dots, \lambda_k$ distinct. We show every vector is a sum of eigenvectors.

The polynomials. For each i let

$$q_i(x) = \prod_{j \neq i} \frac{x - \lambda_j}{\lambda_i - \lambda_j} = \frac{(x-\lambda_1)\cdots\widehat{(x-\lambda_i)}\cdots(x-\lambda_k)}{(\lambda_i-\lambda_1)\cdots\widehat{(\lambda_i-\lambda_i)}\cdots(\lambda_i-\lambda_k)},$$

where the hat means that factor is left out. The denominator is nonzero because the λ_j are distinct. Each q_i has degree $k-1$, and

$$q_i(\lambda_i) = 1, \qquad q_i(\lambda_j) = 0 \ \ (j \neq i).$$

They add up to 1. The polynomial $q_1 + \cdots + q_k - 1$ has degree at most $k-1$, and at each λ_j it takes the value $1 - 1 = 0$. A nonzero polynomial of degree at most $k-1$ has at most $k-1$ roots, so $q_1 + \cdots + q_k - 1 = 0$, that is, $q_1(T) + \cdots + q_k(T) = I$.

Each $q_i(T)$ lands in an eigenspace. Multiplying q_i by the missing factor gives

$$(x - \lambda_i)\, q_i(x) = \frac{m_T(x)}{\prod_{j \neq i}(\lambda_i - \lambda_j)},$$

a constant multiple of m_T. So $(T - \lambda_i I)\, q_i(T) = 0$: for every v, the vector $w_i = q_i(T)v$ satisfies $T w_i = \lambda_i w_i$, so $w_i \in V_{\lambda_i}$.

Conclusion. For every $v \in V$,

$$v = I v = q_1(T)v + \cdots + q_k(T)v = w_1 + \cdots + w_k, \qquad w_i \in V_{\lambda_i}.$$

So $V = V_{\lambda_1} + \cdots + V_{\lambda_k}$, and the sum is direct by theorem 1.2. A basis of each V_{λ_i} together gives an eigenbasis.

For example, if $m_T = (x-a)(x-b)$ with $a \neq b$, then $q_1 = \frac{x-b}{a-b}$, $q_2 = \frac{x-a}{b-a}$, and every v splits as

$$v = \frac{(T - bI)v}{a-b} + \frac{(T - aI)v}{b-a},$$

where the first term is an a-eigenvector (apply $T - aI$ to it and get $\frac{m_T(T)v}{a-b} = 0$) and the second a b-eigenvector.

(d) $(J_{\lambda,d} - \lambda I)^s = 0$ exactly when $s \geq d$, and $(J_{\mu,d} - \lambda I)$ is invertible for $\mu \neq \lambda$ (*Ex*). □

Example 3.3

A projection $P \neq 0, I$ has $m_P = x(x-1)$, so it is diagonalizable. The matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has $m = (x-1)^2$, so it is not.

3.5 Block matrices

Proposition 3.5 (Tutorial 6 Q11). *Let $V = V_1 \oplus \cdots \oplus V_c$ and $W = W_1 \oplus \cdots \oplus W_r$. A linear map $f : V \to W$ is the same as an $r \times c$ matrix (f_{ij}) of linear maps $f_{ij} : V_j \to W_i$: f_{ij} is f restricted to V_j, followed by taking the W_i-component. Addition and composition of linear maps become addition and multiplication of these block matrices, with the usual row-by-column rule (keeping the order of composition) (Ex).*

Proposition 3.6. *For square blocks A ($k \times k$) and D ($\ell \times \ell$):*

$$\det \begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \det A \cdot \det D.$$

Proof. Proof 1 (uniqueness of det*).* Fix B and D, and let $f(A)$ be the left side. As a function of the k columns of A, f is multilinear and alternating (it is det of the big matrix, viewed through its first k columns). By the uniqueness of such functions (Sep 9), $f(A) = \det(A)\, f(I_k)$. At $A = I_k$, column operations (subtract multiples of the first k columns) clear B without changing the determinant, leaving $\det \begin{bmatrix} I & 0 \\ 0 & D \end{bmatrix} = \det D$.

Proof 2 (factor). If A is singular, the first k columns are dependent and both sides are 0. Otherwise

$$\begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} I & A^{-1}B \\ 0 & D \end{bmatrix},$$

and the two factors have determinants $\det A$ and $\det D$ (expand along rows or columns) (*Ex*). □

3.6 Operators on spaces of matrices

These appeared in both Test 1s (2025: $X \mapsto AX - XB$; 2026: $X \mapsto AX$). E_{ij} denotes the matrix with 1 in position (i, j) and 0 elsewhere; these n^2 matrices are a basis of $M_n(\mathbb{F})$.

Proposition 3.7. *Let $A, B \in M_n(\mathbb{F})$, and order the basis of $M_n(\mathbb{F})$ column by column: $E_{11}, E_{21}, \ldots, E_{n1}, E_{12}, \ldots, E_{n2}, \ldots$. Then the coordinate vector of X is its columns stacked on top of each other.*

1. *$L_A(X) = AX$ has matrix* $\operatorname{diag}(A, \ldots, A)$ *(n copies), since the j-th column of AX is A times the j-th column of X. So* $\det L_A = (\det A)^n$, $\operatorname{tr} L_A = n \operatorname{tr} A$, *and $p_{L_A} = p_A^n$.*
2. *$R_B(X) = XB$ has* $\det R_B = (\det B)^n$ *and* $\operatorname{tr} R_B = n \operatorname{tr} B$*: order the basis row by row instead, and use that the i-th row of XB is the i-th row of X times B (Ex).*
3. *$M \mapsto AMB$ is $L_A R_B$, so its determinant is* $(\det A)^n (\det B)^n$. *Its trace is* $\operatorname{tr} A \cdot \operatorname{tr} B$ *(Ex).*
4. *If $Av = \lambda v$ and $B^T w = \mu w$, then $X = vw^T$ satisfies $AX = \lambda X$ and $XB = v(B^T w)^T = \mu X$, so $AX - XB = (\lambda - \mu)X$. If A and B are diagonalizable, with eigenbases $v_1, \ldots, v_n$ of A and $w_1, \ldots, w_n$ of B^T, the n^2 matrices $v_i w_j^T$ form a basis of M_n (Ex), so $X \mapsto AX - XB$ is diagonalizable with eigenvalues $\lambda_i - \mu_j$.*

Remark 3.6.1. In Test 1 the row-by-row ordering made the matrix of L_A interleaved and hid $\operatorname{diag}(A, \ldots, A)$. Choose the ordering that matches how the operator acts, and say which ordering you use. A second route for (a), without blocks: $L_A L_{A'} = L_{AA'}$, so $A \mapsto \det L_A$ is multiplicative, and it suffices to compute $\det L_E$ for elementary matrices E (*Ex*).

Exercises

From the board and the tutorials

3.6.1 *Tutorial 6 Q4* **[2]** Prove proposition 3.1, including (d). Is every projection diagonalizable? What is its trace?

3.6.2 *Tutorial 6 Q7* **[1]** Prove theorem 3.2. Is the converse of (b) true? Extend (b) to an infinite family of commuting diagonalizable operators.

3.6.3 *Tutorial 6 Q8, Q9* **[2]** Write out Proofs 1 and 2 of Cayley–Hamilton in full. In Q9, also decide whether Tutorial 6 Q5(b) remains true with $\mathbb{C}$ replaced by $\mathbb{R}$ everywhere (including similarity over $\mathbb{R}$).

3.6.4 *Tutorial 6 Q11* **[2]** Prove the block matrix proposition: (a) maps $U \to \bigoplus W_i$ correspond to tuples (f_i) (row or column?); (b) maps $\bigoplus V_j \to U$; (c) maps $\bigoplus V_j \to \bigoplus W_i$ correspond to $r \times c$ matrices of maps; (d) ordinary matrices are the special case of 1-dimensional summands; (e) addition and composition become matrix addition and multiplication.

3.6.5 *HW6 Q1* **[2]** Call T a projection if $V = K \oplus R$ with K, R nonzero, T-invariant, $T|_K = 0$ and $T|_R = \mathrm{id}$. Show that then $K = \ker T$ and $R = \operatorname{im} T$, and that T is such a projection if and only if $m_T = x(x-1)$.

3.6.6 *HW6 Q4* **[1]** Let V have basis u, v, w and $T(u) = v$, $T(v) = w$, $T(w) = v$. Over $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{F}_p$, find the minimal polynomial of T, decide whether T is diagonalizable, and if so diagonalize its matrix.

From the books

3.6.7 *Artin 4.5.6* **[2]** Let V have basis $v_0, \dots, v_n$, and define $T(v_i) = v_{i+1}$ for $i < n$ and $T(v_n) = a_0 v_0 + \cdots + a_n v_n$. Find the matrix of T and its characteristic polynomial.

3.6.8 *Artin 4.5.9, 4.5.10, 4.3.4* **[1]** (a) Find the trace and determinant of $M \mapsto AM$ on $m \times m$ matrices. (b) Find the trace and determinant of $M \mapsto AMB$ on $n \times n$ matrices. (c) For a complex $n \times n$ matrix B, is $A \mapsto AB - BA$ always singular?

3.6.9 *Artin 5.2.1, 5.2.2* **[1]** (a) Use Cayley–Hamilton to write A^{-1} in terms of A, $(\det A)^{-1}$ and the coefficients of p_A; check the 2×2 case. (b) Let A be $m \times m$ and B be $n \times n$, and $T(M) = AMB$ on $m \times n$ complex matrices. Build eigenvectors of T from an eigenvector X of A and an eigenvector Y of B^T; find the eigenvalues and the trace of T.

3.6.10 *Artin 5.2.3, 5.2.4* **[1]** (a) For complex A, show that $T(M) = AM - MA$ has rank at most $n^2 - n$, and find its eigenvalues in terms of those of A. (b) Let A, B be diagonalizable. Show that some invertible P makes both $P^{-1}AP$ and $P^{-1}BP$ diagonal if and only if $AB = BA$.

3.6.11 *Axler 5B.3, 5B.11* **[2]** (a) Let T on $\mathbb{F}^n$ ($n > 1$) have the matrix with every entry 1. Find its eigenvalues, eigenvectors and minimal polynomial. (b) For a 2×2 matrix $\begin{bmatrix} a & c \\ b & d \end{bmatrix}$, prove $T^2 - (a+d)T + (ad-bc)I = 0$, and show that the minimal polynomial is $x - a$ if $b = c = 0$ and $a = d$, and $x^2 - (a+d)x + (ad-bc)$ otherwise.

3.6.12 *Axler 5B.7, 5B.8* **[3]** (a) Give S, T on $\mathbb{F}^2$ such that ST and TS have different minimal polynomials. If one of S, T is invertible, prove the minimal polynomials agree. (b) Find the minimal polynomial of the rotation of $\mathbb{R}^2$ by $1°$. (It is not $x^{180} + 1$.)

3.6.13 *Axler 5B.14, 5B.16* **[3]** (a) T has minimal polynomial $4 + 5x - 6x^2 - 7x^3 + 2x^4 + x^5$. Find the minimal polynomial of T^{-1}. (b) Show that the minimal polynomial of the companion matrix (ones just below the diagonal, last column $-a_0, \dots, -a_{n-1}$) is $a_0 + a_1x + \cdots + a_{n-1}x^{n-1} + x^n$. So every monic polynomial is a minimal polynomial.

3.6.14 *Axler 5B.19, 5B.20, 5B.22* **[3]** (a) Show that $\{q(T) : q \text{ a polynomial}\}$ is a subspace of $L(V)$ of dimension $\deg m_T$. (b) If T on $\mathbb{F}^4$ has eigenvalues exactly $3, 5, 8$, prove $(T - 3I)^2(T - 5I)^2(T - 8I)^2 = 0$. (c) Prove that T is invertible if and only if $I \in \operatorname{span}(T, T^2, \dots, T^{\dim V})$.

3.6.15 *Axler 5D.1* **[1]** Let V be a finite-dimensional complex space. (a) If $T^4 = I$, prove T is diagonalizable. (b) If $T^4 = T$, prove T is diagonalizable. (c) Give T on $\mathbb{C}^2$ with $T^4 = T^2$ that is not diagonalizable.

3.6.16 *Axler 5D.4, 5D.5* **[3]** (a) Show that $V = \ker T \oplus \operatorname{im} T$, $V = \ker T + \operatorname{im} T$ and $\ker T \cap \operatorname{im} T = 0$ are equivalent. (b) Over $\mathbb{C}$, prove that T is diagonalizable if and only if $V = \ker(T - \lambda I) \oplus \operatorname{im}(T - \lambda I)$ for every $\lambda \in \mathbb{C}$.

3.6.17 *Axler 5D.15* **[3]** Over $\mathbb{C}$, with $p = m_T$, prove that the following are equivalent: T is diagonalizable; p is not a multiple of $(x-\lambda)^2$ for any λ; p and p' have no common zero; $\gcd(p, p') = 1$. (So diagonalizability can be decided without finding eigenvalues.)

3.6.18 *Axler 5E.1, 5E.3* **[3]** (a) Give commuting S, T on $\mathbb{F}^4$ with a subspace invariant under S but not T, and one invariant under T but not S. (b) If $ST = TS$ and p is a polynomial, prove that $\ker p(S)$ and $\operatorname{im} p(S)$ are T-invariant.

Harder or unusual

3.6.19 *Napkin 10D* **[2]** Let $T\colon \mathbb{C}^2 \to \mathbb{C}^2$ satisfy $T^{2011} = I$. Must T be diagonalizable?

3.6.20 *Axler 5D.14* **[3]** (a) Give T over $\mathbb{C}$ with T^2 diagonalizable but T not. (b) If T is invertible over $\mathbb{C}$ and $k \geq 1$, prove T is diagonalizable if and only if T^k is.

3.6.21 *Axler 5B.15* **[4]** Over $\mathbb{C}$ with $\dim V > 0$, show that $f(\lambda) = \dim \operatorname{im}(T - \lambda I)$ is not a continuous function of λ.

3.6.22 *Axler 5A.43* **[4]** If $\dim V > 1$, prove that $\{p(T) : p \text{ a polynomial}\} \neq L(V)$.

3.6.23 *Axler 5E.6* **[4]** Let S, T commute on a nonzero finite-dimensional complex space. Prove that there are $\alpha, \lambda \in \mathbb{C}$ with $\operatorname{im}(S - \alpha I) + \operatorname{im}(T - \lambda I) \neq V$.

3.6.24 [3] (Commutant.) Find all $n \times n$ complex matrices commuting with $J_{0,n}$, and show they are exactly the polynomials in $J_{0,n}$ (2024 HW4 Q4(ii)). More generally, if T has n distinct eigenvalues, show that every operator commuting with T is a polynomial in T of degree at most $n - 1$ (2024 HW5 Q1).

Fitting Decomposition and Jordan Form

4

The plan of this chapter (Oct 13): split V into pieces on which T has a single eigenvalue (Step 1, via Fitting), split each piece into chains (Step 2), and show the chain lengths are determined by T (Step 3). Only the statements of Fitting, Step 1 and Step 2 are on the test; the proof of Step 3 is.

4.1 Fitting decomposition

The terms used below, collected from Chapter 2:

- A subspace $U \subseteq V$ is *T-stable* (or T-invariant) if $T(U) \subseteq U$. Then the *restriction* $T|_U : U \to U$, $u \mapsto Tu$, is an operator on U (lemma 2.1). Kernels and images of operators that commute with T, such as T^k, are T-stable (Remark in §2.1).
- T is *nilpotent on U* if $T|_U$ is nilpotent, that is, $T^N U = 0$ for some N. T is an *isomorphism on U* if $T|_U : U \to U$ is bijective.
- A chain of subspaces *stabilizes* at step N if all its terms from the N-th on are equal. For kernels this happens as soon as two consecutive terms are equal (lemma 2.3).
- $\ker T^\infty$ and $\operatorname{im} T^\infty$ denote the terms at which the two chains below stabilize; by the theorem, $\ker T^n$ and $\operatorname{im} T^n$ will do.

Fitting decomposition **Theorem 4.1**

The chains

$$0 \subseteq \ker T \subseteq \ker T^2 \subseteq \cdots, \qquad V \supseteq \operatorname{im} T \supseteq \operatorname{im} T^2 \supseteq \cdots$$

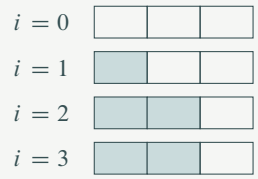
stabilize as soon as a term repeats, so certainly by the n-th term. Write $\ker T^\infty = \ker T^n$ and $\operatorname{im} T^\infty = \operatorname{im} T^n$. Then

$$V = \ker T^\infty \oplus \operatorname{im} T^\infty.$$

Both summands are T-stable, T is nilpotent on $\ker T^\infty$, and T is an isomorphism on $\operatorname{im} T^\infty$. Moreover $\ker T^\infty$ is the largest T-stable subspace on which T is nilpotent, and $\operatorname{im} T^\infty$ is the largest one on which T is an isomorphism.

Example 4.1

Let $T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ on $\mathbb{F}^3$, so $Te_1 = 0$, $Te_2 = e_1$, $Te_3 = 2e_3$. Each bar below has $3 = \dim V$ cells: the shaded ones count $\dim \ker T^i$ and the white ones $\dim \operatorname{im} T^i$ (rank–nullity).



$i = 0$	$\ker T^0 = 0$	$\operatorname{im} T^0 = \mathbb{F}^3$
$i = 1$	$\ker T^1 = \operatorname{span}(e_1)$	$\operatorname{im} T^1 = \operatorname{span}(e_1, e_3)$
$i = 2$	$\ker T^2 = \operatorname{span}(e_1, e_2)$	$\operatorname{im} T^2 = \operatorname{span}(e_3)$
$i = 3$	$\ker T^3 = \operatorname{span}(e_1, e_2)$	$\operatorname{im} T^3 = \operatorname{span}(e_3)$

Both chains stop changing at $i = 2$, and then $\mathbb{F}^3 = \operatorname{span}(e_1, e_2) \oplus \operatorname{span}(e_3)$: T is nilpotent on the first summand and multiplies the second by 2. At $i = 1$ the sum is not direct, since e_1 lies in both $\ker T$ and $\operatorname{im} T$; this is why the theorem uses a high power of T.

Proof (class; not on the test). The kernel chain was handled in lemma 2.3; for images, $\dim \operatorname{im} T^i = n - \dim \ker T^i$ by rank–nullity, so the image chain stabilizes at the same step.

Direct sum. By rank–nullity, $\dim \ker T^n + \dim \operatorname{im} T^n = n$, so it is enough to show $\ker T^n \cap \operatorname{im} T^n = 0$. Let v be in both: $v = T^n w$ and $T^n v = 0$. Then $T^{2n} w = 0$, so $w \in \ker T^{2n} = \ker T^n$, and $v = T^n w = 0$.

The two restrictions. Both subspaces are T-stable since T commutes with T^n. On $\ker T^n$, $T^n = 0$. On $\operatorname{im} T^n$, T maps $\operatorname{im} T^n$ onto $\operatorname{im} T^{n+1} = \operatorname{im} T^n$, so $T|_{\operatorname{im} T^n}$ is surjective, hence bijective.

"Largest". If U is T-stable and $T|_U$ is nilpotent, then $(T|_U)^{\dim U} = 0$, so $U \subseteq \ker T^{\dim U} \subseteq \ker T^n$. If $T|_U$ is an isomorphism, then $U = T^n(U) \subseteq \operatorname{im} T^n$. □

Another proof of $V = \ker T^\infty + \operatorname{im} T^\infty$ (Napkin 9I). Choose N beyond which both chains are constant. Given v, the vector $T^N v$ lies in $\operatorname{im} T^N = \operatorname{im} T^{2N}$, so $T^N v = T^{2N} w$ for some w. Then $v = (v - T^N w) + T^N w$, where $T^N w \in \operatorname{im} T^N$ and $T^N(v - T^N w) = T^N v - T^{2N} w = 0$. This gives the sum without counting dimensions. □

4.2 Generalized eigenspaces

Assume from now on that $p_T(x) = \prod_{i=1}^{r} (x - \lambda_i)^{m_i}$ with the λ_i distinct, so $\sum_i m_i = n$. This is automatic over $\mathbb{C}$.

A *generalized λ-eigenvector* is a v with $(T-\lambda I)^k v = 0$ for some $k \geq 1$. Together with 0 they form the *generalized λ-eigenspace* $V_\lambda^{\text{gen}} = \ker(T-\lambda I)^\infty = \ker(T-\lambda I)^n$, which is T-stable (it is the kernel of a polynomial in T) and contains the ordinary eigenspace V_λ.

Definition 4.1

Step 1
$V = \bigoplus_{i=1}^r V_{\lambda_i}^{\text{gen}}$, and $\dim V_{\lambda_i}^{\text{gen}} = m_i$. In fact $V_{\lambda_i}^{\text{gen}} = \ker(T-\lambda_i I)^{m_i}$.

Theorem 4.2

Proof 1 (class, Oct 13): Fitting and induction on the number of eigenvalues. Apply Fitting to $S = T - \lambda_1 I$: $V = U \oplus W$ with $U = \ker S^\infty = V_{\lambda_1}^{\text{gen}}$ and $W = \operatorname{im} S^\infty$, both T-stable (they are S-stable, and $T = S + \lambda_1 I$).

1. *On U:* $S|_U$ is nilpotent, so by theorem 2.4(B) it is strictly upper triangular in some basis of U, and $T|_U = S|_U + \lambda_1 I$ is upper triangular with every diagonal entry λ_1. So $p_{T|_U} = (x-\lambda_1)^{\dim U}$.
2. *On W:* $S|_W$ is invertible, so λ_1 is not an eigenvalue of $T|_W$, that is, not a root of $p_{T|_W}$.
3. *Compare.* In a basis of U followed by a basis of W, the matrix of T is block diagonal, so $p_T = p_{T|_U}\, p_{T|_W}$. Comparing with $p_T = (x-\lambda_1)^{m_1}\prod_{i\geq 2}(x-\lambda_i)^{m_i}$ gives $\dim U = m_1$ and $p_{T|_W} = \prod_{i\geq 2}(x-\lambda_i)^{m_i}$.
4. *Induct.* Apply the same argument to $T|_W$, which has the $r-1$ eigenvalues $\lambda_2, \ldots, \lambda_r$. This needs $V_{\lambda_i}^{\text{gen}}(T) = V_{\lambda_i}^{\text{gen}}(T|_W)$ for $i \geq 2$, that is, $V_{\lambda_i}^{\text{gen}} \subseteq W$: on U the operator $T - \lambda_i I = S + (\lambda_1 - \lambda_i)I$ is invertible (nilpotent plus nonzero scalar), so it kills no nonzero vector of U in any power (*Ex*).

Since $(T-\lambda_1 I)^{m_1}$ already kills U (a nilpotent operator on an m_1-dimensional space has m_1-th power 0), $V_{\lambda_1}^{\text{gen}} = \ker(T-\lambda_1 I)^{m_1}$. □

Proof 2 (board: "gcd of polynomials"): Bézout. Lemma. If $p_T = fg$ with $\gcd(f,g) = 1$, then $V = \ker f(T) \oplus \ker g(T)$. Indeed, the Euclidean algorithm for polynomials gives a, b with $af + bg = 1$, so for $v \in V$

$$v = a(T)f(T)v + b(T)g(T)v.$$

The first term is killed by $g(T)$, since $g(T)a(T)f(T) = a(T)p_T(T) = 0$ by Cayley–Hamilton; so it lies in $\ker g(T)$. Likewise the second lies in $\ker f(T)$. If $v \in \ker f(T) \cap \ker g(T)$, both terms vanish and $v = 0$.

Apply it. Take $f = (x-\lambda_1)^{m_1}$ and $g = p_T/f$, which are coprime. Then $V = \ker(T-\lambda_1 I)^{m_1} \oplus \ker g(T)$, both T-stable. On the first summand T has only the

eigenvalue λ_1; on the second, λ_1 is not an eigenvalue (an eigenvector there would satisfy $g(\lambda_1)v = 0$, but $g(\lambda_1) \neq 0$). The characteristic polynomials of the two restrictions multiply to p_T, so they are $(x-\lambda_1)^{m_1}$ and g, and the dimensions are m_1 and $n - m_1$. Induct on g.

This proof also shows that the projections onto the summands are polynomials in T, namely $b(T)g(T)$ and $a(T)f(T)$. □

4.3 Nilpotent Jordan form (Step 2)

Definition 4.2

The *Jordan block* $J_{\lambda,d}$ is the $d \times d$ matrix with λ on the diagonal, 1 just above it, and 0 elsewhere. Let N be nilpotent on W. A *chain* (of length t) is a list $u, Nu, N^2u, \ldots, N^{t-1}u$ of nonzero vectors with $N^t u = 0$; on its span, in the basis $N^{t-1}u, \ldots, Nu, u$, N has matrix $J_{0,t}$ (the "staircase"). A *Jordan basis* for N is a basis of W that is a disjoint union of chains.

A nilpotent operator is usually *not* a single block $J_{0,n}$. In a Jordan basis its matrix is block diagonal, with one block $J_{0,t}$ for each chain of length t. For example, $N = 0$ on $\mathbb{F}^2$ has the Jordan basis e_1, e_2, made of two chains of length 1, and its matrix is $J_{0,1} \oplus J_{0,1}$, the zero matrix; it is not similar to $J_{0,2}$, which has rank 1. The single block $J_{0,n}$ is the case of one chain, which happens exactly when $N^{n-1} \neq 0$ (*Ex*).

For example, here is $N = J_{0,3} \oplus J_{0,2} \oplus J_{0,1}$ on $\mathbb{F}^6$; column j is Ne_j, and the shaded squares are the blocks. Reading the columns: $Ne_1 = 0$, $Ne_2 = e_1$, $Ne_3 = e_2$, $Ne_4 = 0$, $Ne_5 = e_4$, $Ne_6 = 0$.

$$N = \begin{array}{c} \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \end{matrix} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

So the chains are $e_3 \mapsto e_2 \mapsto e_1 \mapsto 0$, $e_5 \mapsto e_4 \mapsto 0$ and $e_6 \mapsto 0$, one for each block, and the length of each chain is the size of its block.

Theorem 4.3

Step 2
Every nilpotent N on a finite-dimensional space W has a Jordan basis. Equivalently, $W = \bigoplus_j W_j$ with each W_j N-stable and $N|_{W_j}$ a single staircase.

Proof 1 (class, Oct 14; Napkin §10.7): induction on the image. Induction on $\dim W$. If $N = 0$, any basis is a Jordan basis (chains of length 1). Otherwise $0 \neq \operatorname{im} N \neq W$: the image is a proper subspace because a nilpotent operator is not surjective.

Step (i): chains in the image. $\operatorname{im} N$ is N-stable and $N|_{\operatorname{im} N}$ is nilpotent, so by induction $\operatorname{im} N$ has a Jordan basis. Every vector of $\operatorname{im} N$ has the form Nv, so we can write the chains of this basis as

$$Nv_i,\ N^2v_i,\ \ldots,\ N^{k_i}v_i \qquad (i=1,\ldots,\ell), \qquad N^{k_i+1}v_i = 0,$$

for some vectors $v_1,\ldots,v_\ell \in W$.

Step (ii): lengthen each chain by one. Prepend v_i to the i-th chain: $v_i, Nv_i, \ldots, N^{k_i}v_i$ is again a chain.

Step (iii): add kernel vectors. The last vectors $N^{k_1}v_1,\ldots,N^{k_\ell}v_\ell$ lie in $\ker N$ and are independent (they are part of a basis). Extend them to a basis of $\ker N$ by vectors $v_{\ell+1},\ldots,v_m$, where $m = \dim\ker N$; each of these is a chain of length 1.

For example, for $N = J_{0,3}\oplus J_{0,2}$ on $\mathbb{F}^5$ (chains $e_3\mapsto e_2\mapsto e_1$ and $e_5\mapsto e_4$), $\operatorname{im} N = \operatorname{span}(e_1,e_2,e_4)$ has the chains $e_2\mapsto e_1$ and e_4. Step (ii) puts e_3 before the first and e_5 before the second, and step (iii) adds nothing, since e_1,e_4 already span $\ker N$. For $N = J_{0,2}\oplus J_{0,1}$, step (iii) would add e_3.

Step (iv): check that $\mathcal{B}$ (all vectors of steps (ii) and (iii)) is a basis.

1. Count: $|\mathcal{B}| = \underbrace{\ell}_{\text{new } v_i} + \underbrace{\dim\operatorname{im} N}_{\text{old chains}} + \underbrace{(m-\ell)}_{\text{kernel vectors}} = \dim\operatorname{im} N + \dim\ker N = \dim W$.
2. Independence: suppose a linear combination of $\mathcal{B}$ is 0. Apply N. The kernel vectors $v_{\ell+1},\ldots,v_m$ and the chain ends $N^{k_i}v_i$ go to 0, and every other vector moves one step along its chain, so we get a combination of the Jordan basis of $\operatorname{im} N$ with the same coefficients as those of $v_i,\ldots,N^{k_i-1}v_i$. Those coefficients are therefore 0. What remains is a combination of $N^{k_1}v_1,\ldots,N^{k_\ell}v_\ell, v_{\ell+1},\ldots,v_m$, a basis of $\ker N$, so the remaining coefficients are 0 too (*Ex*).

So $\mathcal{B}$ is a Jordan basis. □

Proof 2: from the top of the kernel chain down. Let q be the index of N ($N^q = 0 \neq N^{q-1}$) and $K_i = \ker N^i$, so $0 = K_0 \subsetneq K_1 \subsetneq \cdots \subsetneq K_q = W$ (lemma 2.3). Put $d_i = \dim K_i$, so $d_0 = 0$ and $d_q = \dim W$.

Working modulo K_{i-1}. Fix $i \geq 1$ and a list $x_1,\ldots,x_s$ of vectors of K_i.

- The list is *independent modulo K_{i-1}* if $\sum_j a_j x_j \in K_{i-1}$ implies $a_1 = \cdots = a_s = 0$. Equivalently, the x_j are independent and $\operatorname{span}(x_1,\ldots,x_s)\cap K_{i-1} = 0$ (*Ex*).
- The list is a *basis of K_i modulo K_{i-1}* if it is independent modulo K_{i-1} and $K_i = \operatorname{span}(x_1,\ldots,x_s) + K_{i-1}$. Equivalently, $K_i = \operatorname{span}(x_1,\ldots,x_s)\oplus K_{i-1}$ with the x_j a basis of the first summand; so $s = d_i - d_{i-1}$, and a basis of K_{i-1} together with the x_j is a basis of K_i.

For example, if $W = \mathbb{F}^2$ and $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, then $K_1 = \mathbb{F}e_1$, $K_2 = \mathbb{F}^2$, and the single vector $e_2 + 5e_1$ is a basis of K_2 modulo K_1, while e_1 is not even independent modulo K_1.

Three facts.

(a) *Extension.* A list in K_i that is independent modulo K_{i-1} extends to a basis of K_i modulo K_{i-1}. (Join it to a basis of K_{i-1}; the result is independent. Extend it to a basis of K_i and keep everything except the basis of K_{i-1} (*Ex*).)

(b) $N(K_{i+1}) \subseteq K_i$, since $N^i(Nx) = N^{i+1}x = 0$.

(c) *N pushes independence down a level.* If $x_1, \dots, x_s \in K_{i+1}$ are independent modulo K_i and $i \geq 1$, then $Nx_1, \dots, Nx_s \in K_i$ are independent modulo K_{i-1}. Indeed, if $\sum_j a_j Nx_j \in K_{i-1}$, then $N^{i-1}N\left(\sum_j a_j x_j\right) = 0$, so $\sum_j a_j x_j \in K_i$, so all $a_j = 0$.

Construction. We choose lists $B_q, B_{q-1}, \dots, B_1$, where B_i is a basis of K_i modulo K_{i-1}, from the top down.

- *Level q.* Let B_q be any basis of K_q modulo K_{q-1}, for instance a basis of a complement of K_{q-1} in W.
- *Level $i < q$.* The list $N(B_{i+1})$ lies in K_i by (b) and is independent modulo K_{i-1} by (c). By (a), extend it to a basis B_i of K_i modulo K_{i-1}. Call the added vectors *new at level i*; all of B_q counts as new at level q.

For example, for $N = J_{0,3} \oplus J_{0,2} \oplus J_{0,1}$ on $\mathbb{F}^6$ (chains $e_3 \mapsto e_2 \mapsto e_1$, $e_5 \mapsto e_4$, e_6) we have $K_1 = \operatorname{span}(e_1, e_4, e_6)$, $K_2 = K_1 + \operatorname{span}(e_2, e_5)$, $K_3 = \mathbb{F}^6$. One run of the construction: $B_3 = (e_3)$; then $B_2 = (Ne_3)$ plus the new vector e_5, that is (e_2, e_5); then $B_1 = (Ne_2, Ne_5)$ plus the new vector e_6, that is (e_1, e_4, e_6).

$B = B_1 \cup \cdots \cup B_q$ is a basis of W. By induction on i, $B_1 \cup \cdots \cup B_i$ is a basis of K_i: this holds for $i = 1$ since $K_0 = 0$, and the inductive step is the last sentence of the definition of a basis modulo K_{i-1}. Take $i = q$.

B is a Jordan basis. If x is new at level i, then $x, Nx, \dots, N^{i-1}x$ lie in $B_i, B_{i-1}, \dots, B_1$ by construction, and $N^i x = 0$ since $x \in K_i$. So x starts a chain of length i inside B. Every $y \in B_j$ lies on exactly one such chain: either y is new at level j, or $y = Nz$ for a unique $z \in B_{j+1}$ (the vectors of $N(B_{j+1})$ are distinct, being independent), and we repeat with z. So B is a disjoint union of chains.

Step 3 for free. Each chain of length $\geq i$ has exactly one vector in B_i, and every vector of B_i lies on such a chain. So the number of chains of length $\geq i$ is $|B_i| = d_i - d_{i-1}$. In the example $|B_3|, |B_2|, |B_1| = 1, 2, 3$, and the numbers of chains of length exactly $3, 2, 1$ are $1, 2-1 = 1$ and $3-2 = 1$. □

Proof 3: a longest chain splits off (Halmos §57). Induction on $\dim W$; if $W = 0$ there is nothing to prove. Let q be the index of N and pick x with $N^{q-1}x \neq 0$. Then

$U = \operatorname{span}(x, Nx, \dots, N^{q-1}x)$ is an N-stable subspace of dimension q, spanned by one chain (Axler 8A.2). We find an N-stable complement of U.

The functional. Extend $N^{q-1}x$ to a basis of W and let $\varphi\colon W \to \mathbb{F}$ be the linear functional taking the coefficient of $N^{q-1}x$ in this basis. Then $\varphi(N^{q-1}x) = 1$. Define the linear map

$$\Phi\colon W \to \mathbb{F}^q, \qquad \Phi(v) = \big(\varphi(v), \varphi(Nv), \dots, \varphi(N^{q-1}v)\big),$$

and let $C = \ker \Phi = \{v \in W : \varphi(N^k v) = 0 \text{ for } k = 0, 1, \dots, q-1\}$.

1. C is N-stable: if $v \in C$, then $\varphi(N^k(Nv)) = \varphi(N^{k+1}v) = 0$ for $k \le q-2$, and for $k = q-1$ it is $\varphi(N^q v) = \varphi(0) = 0$.
2. $C \cap U = 0$: a nonzero $u \in U$ is $u = \sum_{i \ge j} a_i N^i x$ with $a_j \neq 0$; then $N^{q-1-j}u = a_j N^{q-1}x$ (higher powers vanish), so $\varphi(N^{q-1-j}u) = a_j \neq 0$ and $u \notin C$.
3. $\dim C \ge \dim W - q$: by rank–nullity, $\dim C = \dim W - \operatorname{rank}\Phi$, and $\operatorname{rank}\Phi \le \dim \mathbb{F}^q = q$.

By (2), $U + C$ is direct, so $q + \dim C = \dim(U \oplus C) \le \dim W$. With (3), $\dim C = \dim W - q$, hence $W = U \oplus C$.

Induction. C is N-stable and $N|_C$ is nilpotent, and $\dim C < \dim W$ because $q \ge 1$. By induction C has a Jordan basis for $N|_C$. Adjoining the chain $x, Nx, \dots, N^{q-1}x$ gives a basis of $U \oplus C = W$ made of chains, a Jordan basis for N. □

Theorem 4.4

Jordan form

If p_T factors fully, V has a basis in which the matrix of T is block diagonal with Jordan blocks. The blocks for λ_i come from a Jordan basis of $N_i = (T - \lambda_i I)|_{V_{\lambda_i}^{\mathrm{gen}}}$, and their sizes add up to m_i. For each λ_i, the multiset of block sizes is uniquely determined by T (Step 3). So two matrices in Jordan form are similar if and only if they have the same multiset of block sizes for every eigenvalue.

Proof of existence. Step 1 splits V into the generalized eigenspaces (the "macro" decomposition). On $V_{\lambda_i}^{\mathrm{gen}}$, N_i is nilpotent, so Step 2 gives a Jordan basis for N_i (the "micro" decomposition). On the span of a chain, N_i has matrix $J_{0,t}$, so $T = N_i + \lambda_i I$ has matrix $J_{\lambda_i,t}$. Put all these bases together. □

Another proof of existence (Napkin §10.6): indecomposable pieces. Call T *indecomposable* on V if V is not a direct sum of two nonzero T-stable subspaces. Any V is a direct sum of T-stable pieces on each of which T is indecomposable: keep splitting; the dimensions drop, so this stops. So assume T is indecomposable, and over $\mathbb{C}$ let λ be an eigenvalue. Fitting for $S = T - \lambda I$ gives $V = \ker S^\infty \oplus \operatorname{im} S^\infty$, both T-stable, with $\ker S^\infty \neq 0$ (it contains an eigenvector). Indecomposability forces $\operatorname{im} S^\infty = 0$, so S is nilpotent. By Step 2, V is a direct sum of chains for S, and indecomposability forces a single chain. So T is a single Jordan block. □

4.4 Uniqueness (Step 3)

What Step 3 says. A nilpotent N has many Jordan bases. For example, for $N = 0$ on $\mathbb{F}^2$ every basis is a Jordan basis. Different Jordan bases might, a priori, give different block sizes. Step 3 says they cannot: the number of blocks of each size is computed from the numbers $\dim\ker N^i$, which do not mention any basis.

Powers of one block. Each power of $J_{0,j}$ moves the line of 1s one step further up, and $J_{0,j}^i = 0$ once $i \geq j$. For $j = 3$:

$$J_{0,3} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \qquad J_{0,3}^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad J_{0,3}^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The nonzero columns of $J_{0,j}^i$ are distinct standard basis vectors, so they are independent, and the kernel is spanned by the e_m whose column is zero. Here there are 1, 2, 3 zero columns, so $\dim\ker J_{0,3}^i = \min(i, 3)$. In general $\dim\ker J_{0,j}^i = \min(i, j)$.

Several blocks. The powers of $N = J_{0,3} \oplus J_{0,2} \oplus J_{0,2} \oplus J_{0,1}$ (on $\mathbb{F}^8$) are block diagonal with the powers of the blocks:

$$N = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad N^2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad N^3 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Again the nonzero columns are distinct standard basis vectors, so $d_i = \dim\ker N^i$ is the number of zero columns, and it adds up block by block: $d_1 = 1 + 1 + 1 + 1 = 4$, $d_2 = 2 + 2 + 2 + 1 = 7$, $d_3 = 3 + 2 + 2 + 1 = 8$. Each block of size $\geq i$ adds exactly 1 when the power goes from $i - 1$ to i, and the smaller blocks add 0:

i	1	2	3	4
$d_i = \dim\ker N^i$	4	7	8	8
$d_i - d_{i-1} =$ blocks of size $\geq i$	4	3	1	0
$k_i =$ blocks of size exactly i	1	2	1	0

The last row is the difference of consecutive entries of the row above it: $4 - 3 = 1$, $3 - 1 = 2$, $1 - 0 = 1$.

Step 3, proof required — **Theorem 4.5**

Let N be nilpotent on W. Fix a Jordan basis, and let k_j be the number of its chains of length j, that is, the number of blocks $J_{0,j}$ in the matrix. Put $d_i = \dim\ker N^i$,

Theorem 4.5 *continued*

with $d_0 = 0$. Then for every $i \geq 1$

$$d_i = \sum_j \min(i, j)\, k_j,$$
$$d_i - d_{i-1} = k_i + k_{i+1} + \cdots = \#\{\text{blocks of size} \geq i\},$$
$$k_i = (d_i - d_{i-1}) - (d_{i+1} - d_i) = 2d_i - d_{i-1} - d_{i+1}.$$

The d_i are defined without any basis, so the k_i are the same for every Jordan basis of N.

Proof 1 (class, Oct 14). One block. On a single chain $e_j \mapsto e_{j-1} \mapsto \cdots \mapsto e_1 \mapsto 0$ (the basis of $J_{0,j}$), N^i sends e_m to e_{m-i} if $m > i$ and to 0 if $m \leq i$. So for $v = \sum_m a_m e_m$,

$$N^i v = \sum_{m>i} a_m e_{m-i},$$

a combination of *distinct* basis vectors, which is 0 exactly when $a_m = 0$ for all $m > i$. Hence $\ker J_{0,j}^i = \operatorname{span}(e_1, \ldots, e_{\min(i,j)})$ and $\dim \ker J_{0,j}^i = \min(i, j)$.

Kernels of block diagonal matrices add up. The matrix of N is $M = \operatorname{diag}(J_{0,j_1}, \ldots, J_{0,j_s})$, and $M^i = \operatorname{diag}(J_{0,j_1}^i, \ldots, J_{0,j_s}^i)$. Split a vector x into pieces $x_1, \ldots, x_s$ matching the blocks. Then $M^i x = 0$ exactly when $J_{0,j_r}^i x_r = 0$ for every r, so $\ker M^i$ is the direct sum of the kernels of the blocks and

$$d_i = \sum_{r=1}^{s} \min(i, j_r) = \sum_j \min(i, j)\, k_j,$$

grouping the blocks by size.

Solve for the k_j. Written out,

$$d_1 = k_1 + k_2 + k_3 + \cdots,$$
$$d_2 = k_1 + 2(k_2 + k_3 + \cdots),$$
$$d_3 = k_1 + 2k_2 + 3(k_3 + k_4 + \cdots),$$

and so on. Subtracting consecutive lines gives $d_i - d_{i-1} = k_i + k_{i+1} + \cdots$. Subtracting two consecutive such equations gives k_i.

General T. Apply this to $N = (T - \lambda I)|_{V_\lambda^{\text{gen}}}$. Since $\ker(T - \lambda I)^i \subseteq V_\lambda^{\text{gen}}$, the numbers $d_i = \dim \ker(T - \lambda I)^i$ can be computed on all of V. □

Proof 2: ranks. Let $r_i = \operatorname{rank} N^i = \dim W - d_i$. For one block, $\operatorname{rank} J_{0,j}^i = \max(j - i, 0)$, so $r_{i-1} - r_i$ counts the blocks of size $\geq i$: each such block loses exactly one from its rank when the power goes from $i - 1$ to i, and the others contribute 0 (*Ex*). □

Proof 3: induction through the image. The number of blocks is the number of chains, which is $\dim \ker N$, because the last vectors of the chains form a basis of $\ker N$ (Axler 8C.13). Deleting the first vector of every chain gives a Jordan basis of $\operatorname{im} N$ for

$N|_{\operatorname{im} N}$, so the block sizes of $N|_{\operatorname{im} N}$ are those of N minus one, with blocks of size 1 disappearing. By induction on $\dim W$, the block sizes of $N|_{\operatorname{im} N}$ are determined by N; together with the number of blocks, this determines those of N (*Ex*). □

Remark 4.4.1 (Young diagrams). Draw a row of j boxes for each block of size j, longest first; for the sizes $3,2,2,1$ above, the rows have $3,2,2,1$ boxes and the columns have $4,3,1$ boxes, which are the numbers $d_i - d_{i-1}$. The row lengths (the block sizes) and the column lengths (the kernel data) are *conjugate partitions* of $\dim W$, and either determines the other. So nilpotent $n \times n$ matrices up to similarity correspond to partitions of n; for $n = 3$ these are 3, $2+1$, $1+1+1$, that is, $J_{0,3}$, $J_{0,2} \oplus J_{0,1}$ and 0.

Example 4.2

Let $A = \begin{bmatrix} 3 & 1 & 1 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, so $p_A = (x-3)^3$. Then $N = A - 3I = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ has rank 1, so $d_1 = 2$. Since $N^2 = 0$, $d_2 = 3$. There are $d_1 = 2$ blocks, and $d_2 - d_1 = 1$ of them has size at least 2. So the Jordan form is $J_{3,2} \oplus J_{3,1}$.

To find a Jordan basis (Proof 2 above): level 2 needs one vector of $K_2 = \mathbb{F}^3$ outside $K_1 = \ker N$, say $u = (0,1,0)^T$, giving the chain $u \mapsto Nu = (1,0,0)^T \mapsto 0$. Level 1 needs one more kernel vector independent of Nu, such as $(0,1,-1)^T$. With $P = [Nu \;\; u \;\; (0,1,-1)^T]$, $P^{-1}AP = J_{3,2} \oplus J_{3,1}$ (*Ex*).

Remark 4.4.2. The numbers d_i give a test for similarity without writing a Jordan form, which suits the "do not use the Jordan form" rule. If $A \sim B$, then $\dim \ker(A - \lambda I)^i = \dim \ker(B - \lambda I)^i$ for all λ and i, because $(P^{-1}AP - \lambda I)^i = P^{-1}(A - \lambda I)^i P$. With the Jordan form, the converse also holds.

Exercises

From the board and the tutorials

4.4.1 *Oct 14 board* **[1]** Fill in every (*Ex*) in the proof of theorem 4.5. Then do Proofs 2 and 3.

4.4.2 *Oct 14 board, Napkin 10.7.1–10.7.2* **[3]** Finish Proof 1 of theorem 4.3: the count, and the independence (assume a dependence and apply N).

4.4.3 *Oct 8 board* **[3]** In Step 1, can one take $\infty = m_j$ for λ_j? Why is the factorization of p_T unique? Write a full proof of the Fitting decomposition: show spanning explicitly, and show that $T|_{\operatorname{im} T^\infty}$ is bijective.

4.4.4 *Tutorial 6 Q10* **[2]** Suppose $p_T = \prod(x-\lambda_i)^{m_i}$ with distinct λ_i. Are the λ_i and m_i unique? Show $\dim\ker(T-\lambda_i I) \le m_i$ and $\dim\ker(T-\lambda_i I)^\infty = m_i$, that the generalized eigenspaces for distinct λ_i are independent, and that $V = \bigoplus_i \ker(T-\lambda_i I)^{m_i}$. Is the first claim true if p_T does not factor fully?

4.4.5 *HW6 Q5, Artin 4.7.6, 4.4.6* **[2]** Find all invariant subspaces of an operator whose Jordan form is a single block. Compare with the D-invariant subspaces of the polynomials of degree at most n (Artin 4.4.6). Is it the same?

4.4.6 *HW6 Q6* **[1]** Show that there is exactly one similarity class of complex nilpotent N with $\dim\ker N = 4$, $\dim\ker N^2 = 8$, $\dim\ker N^3 = \dim\ker N^4 = 11$. Does the answer change over another field? How many classes are there if 11 is replaced by 13?

4.4.7 *HW6 Q7* **[1]** The minimal polynomial of T is $(t-a)^3(t-b)^4$. Write the generalized eigenspace decomposition. What are the possible values of $\dim V$? For which $\dim V$ is the Jordan form decided by this information? How many Jordan forms are possible when $\dim V = 10$?

From the books

4.4.8 *Artin 4.7.1, 4.7.2* **[1]** (a) Find the Jordan form of $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. (b) Show that $A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$ satisfies $A^2 = A$, and find its Jordan form.

4.4.9 *Artin 4.7.3, 4.7.4, 4.7.5* **[1]** (a) $\dim V = 5$, $p_T = (t-\lambda)^5$ and $\operatorname{rank}(T-\lambda I) = 2$. Find the possible Jordan forms. (b) Find all Jordan forms with characteristic polynomial $(t+2)^2(t-5)^3$; then those where the (-2)-eigenspace is 1-dimensional and the 5-eigenspace is 2-dimensional. (c) What is the Jordan form of a matrix all of whose eigenvectors are multiples of one vector?

4.4.10 *Artin 4.7.7, 4.7.8, 4.7.9* **[1]** (a) Is every complex A with $A^2 = A$ diagonalizable? (b) Is every complex square matrix similar to its transpose? (Halmos §58.8 asks the same.) (c) Find a 2×2 matrix over $\mathbb{F}_p$ that has a power equal to I and an eigenvalue in $\mathbb{F}_p$, but is not diagonalizable.

4.4.11 *Axler 8B.4, 8B.5, 8B.7* **[2]** (a) If $\dim V \geq 2$ and $\ker T^{\dim V-2} \neq \ker T^{\dim V-1}$, prove that T has at most two distinct eigenvalues. (b) If 3 and 8 are eigenvalues of T and $n = \dim V$, prove $V = \ker T^{n-2} \oplus \operatorname{im} T^{n-2}$. (c) If λ has multiplicity d, prove $V_\lambda^{\text{gen}} = \ker(T - \lambda I)^d$.

4.4.12 *Axler 8B.12–8B.16* **[2]** Give operators on $\mathbb{C}^4$ with: (a) $p_T = (x-1)(x-5)^3$ and $m_T = (x-1)(x-5)^2$; (b) $p_T = m_T = x(x-1)^2(x-3)$; (c) $p_T = x(x-1)^2(x-3)$ and $m_T = x(x-1)(x-3)$. Find p_T and m_T for $T(z_1, z_2, z_3, z_4) = (0, z_1, z_2, z_3)$ on $\mathbb{C}^4$ and for $T(z_1, \ldots, z_6) = (0, z_1, z_2, 0, z_4, 0)$ on $\mathbb{C}^6$.

4.4.13 *Axler 8B.18* **[3]** For an eigenvalue λ of T, explain why these four numbers are equal: the exponent of $x - \lambda$ in m_T; the least m with $(T - \lambda I)^m|_{V_\lambda^{\text{gen}}} = 0$; the least m with $\ker(T-\lambda I)^m = \ker(T-\lambda I)^{m+1}$; the least m with $\operatorname{im}(T-\lambda I)^m = \operatorname{im}(T-\lambda I)^{m+1}$.

4.4.14 *Axler 8B.21* **[3]** If p, q are monic complex polynomials with the same zeros and p divides q, prove there is an operator on $\mathbb{C}^{\deg q}$ with characteristic polynomial q and minimal polynomial p.

4.4.15 *Axler 8C.5, 8C.6* **[2]** (a) Find a Jordan basis for $T(w, z) = (-w-z,\ 9w+5z)$ on $\mathbb{C}^2$. (b) Find a Jordan basis of the polynomials of degree at most 4 for the differentiation operator.

4.4.16 *Axler 8C.13, 8C.14* **[2]** (a) If chains $N^{m_k} v_k, \ldots, N v_k, v_k$ $(k = 1, \ldots, s)$ form a basis and $N^{m_k+1} v_k = 0$, prove that $N^{m_1} v_1, \ldots, N^{m_s} v_s$ is a basis of $\ker N$. (So the number of blocks is $\dim \ker N$.) (b) Over $\mathbb{C}$, prove that V is not a direct sum of two nonzero T-invariant subspaces if and only if $m_T = (x - \lambda)^{\dim V}$ for some λ.

4.4.17 *Halmos §58.2, §58.4, §58.6* **[2]** (a) How many similarity classes of 3×3 complex matrices have characteristic polynomial $(x-1)^3$? (b) With ω a primitive cube root of 1, show that $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ is similar to $\operatorname{diag}(1, \omega, \omega^2)$. Generalize to $n\times n$. (c) Show that the 3×3 matrix of all 1s is similar over $\mathbb{C}$ to $\operatorname{diag}(3, 0, 0)$. Generalize.

4.4.18 *Halmos §57.3* **[2]** For nilpotent A of index q, prove $\dim \ker A^{k+1} + \dim \ker A^{k-1} \leq 2 \dim \ker A^k$ for $1 \leq k \leq q-1$. (So the sequence d_i is concave; this is the inequality behind "the number of blocks of size $\geq i$ decreases".)

Harder or unusual

4.4.19 *Halmos §58.9* **[3]** If $\begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix}$ and $\begin{bmatrix} B & 0 \\ 0 & B \end{bmatrix}$ are similar, prove that A and B are similar.

4.4.20 *Halmos §58.7* **[4]** If two real matrices are similar over $\mathbb{C}$, prove that they are similar over $\mathbb{R}$.

4.4.21 *Halmos §57.4, Axler 8B.9* **[3]** (Jordan–Chevalley.) Over $\mathbb{C}$, show that $T = D + N$ with D diagonalizable, N nilpotent and $DN = ND$, and that D and N are unique. (Hint for uniqueness: D and N are polynomials in T, by Proof 2 of Step 1.)

4.4.22 *Axler 8C.1, 8C.3, Halmos §58.3* **[3]** (a) Prove that $T(z_1, z_2, z_3) = (z_2, z_3, 0)$ on $\mathbb{C}^3$ has no square root. (b) Prove that every invertible operator on a complex space has a square root and a cube root. (Hint: $\sqrt{1+x}$ as a power series that stops because N is nilpotent.)

4.4.23 *Axler 8C.4* **[4]** Let V be a real vector space. Prove that $-I$ has a square root if and only if $\dim V$ is even.

4.4.24 *Artin 4.M.7* **[4]** For an operator T on finite-dimensional V, let $K_r = \ker T^r$ and $W_r = \operatorname{im} T^r$. Find all implications among: (1) $K_r = K_{r+1}$; (2) $W_r = W_{r+1}$; (3) $W_r \cap K_1 = 0$; (4) $W_1 + K_r = V$. What changes when V is infinite-dimensional?

4.4.25 **[2]** Show that two nilpotent $n \times n$ matrices are similar if and only if $\operatorname{rank} N^i = \operatorname{rank} M^i$ for all i, and that the number of similarity classes of nilpotent $n \times n$ matrices is the number of partitions of n. Find the smallest n for which two nilpotent matrices with the same rank and the same rank of the square are not similar.

Part 2



Inner Product Spaces

CHAPTER 5	Euclidean Spaces	42
CHAPTER 6	Hermitian Spaces	52
CHAPTER 7	Orthogonal and Unitary Operators	55

5 Euclidean Spaces

5.1 The dot product and inner products

The model is the dot product on $\mathbb{R}^n$: $v \cdot w = \sum_i v_i w_i = v^T w$. It gives

- the *length* or *norm* $\|v\| = \sqrt{v \cdot v}$;
- *orthogonality*: $v \perp w \iff v \cdot w = 0$;
- the *angle* θ between $v, w \neq 0$, by $\cos\theta = \dfrac{v \cdot w}{\|v\|\|w\|}$.

The dot product is (1) bilinear, (2) symmetric and (3) positive definite: $v \cdot v \geq 0$, with equality if and only if $v = 0$.

Definition 5.1

An *inner product* on a real vector space V is a function $\langle \cdot, \cdot \rangle \colon V \times V \to \mathbb{R}$ that is

1. bilinear: $\langle av + v', w \rangle = a\langle v, w \rangle + \langle v', w \rangle$, and similarly in the second slot;
2. symmetric: $\langle v, w \rangle = \langle w, v \rangle$;
3. positive definite: $\langle v, v \rangle > 0$ for $v \neq 0$.

A finite-dimensional real vector space with an inner product is a *Euclidean space*. We define $\|v\| = \sqrt{\langle v, v \rangle}$, $v \perp w \iff \langle v, w \rangle = 0$, and, after proving Cauchy–Schwarz, $\cos\theta = \langle v, w \rangle / (\|v\|\|w\|)$ for $v, w \neq 0$.

Example 5.1

(0) $\mathbb{R}^n$ with the dot product.

(1) Any subspace of an inner product space, with the restricted inner product. (Simple, but useful.)

(2) $C[-1, 1]$, or its subspace of polynomials, with $\langle f, g \rangle = \int_{-1}^{1} f(x)g(x)\,dx$, or with $\int_{-1}^{1} f(x)g(x)w(x)\,dx$ for a positive weight w.

Example 5.1 *continued*

(3) $M_n(\mathbb{R})$ with $\langle A, B\rangle = \operatorname{tr}(A^T B) = \sum_{i,j} a_{ij} b_{ij}$.

(4) $\mathbb{R}^n$ with $\langle x, y\rangle = x^T G y$, for any symmetric G with $x^T G x > 0$ for $x \neq 0$.

Remark 5.1.1. Positive definiteness implies *nondegeneracy*: if $\langle v, x\rangle = 0$ for every $x \in V$, then $v = 0$ (take $x = v$).

5.2 Orthonormal bases

A basis $e_1, \ldots, e_n$ of a Euclidean space is *orthonormal* (ON) if $\langle e_i, e_j\rangle = \delta_{ij}$, that is, $\|e_i\| = 1$ and $e_i \perp e_j$ for $i \neq j$.

Definition 5.2

Proposition 5.1.

1. *Any set of mutually orthogonal nonzero vectors is linearly independent.*
2. *Coordinates in an ON basis are easy to find:* $v = \sum_j \langle e_j, v\rangle\, e_j$.
3. *(Parseval) In an ON basis,* $\langle v, w\rangle = \sum_j \langle e_j, v\rangle\langle e_j, w\rangle$ *and* $\|v\|^2 = \sum_j \langle e_j, v\rangle^2$.

Proof. (a) If $\sum_j c_j w_j = 0$, take the inner product with w_i: all terms with $j \neq i$ vanish, leaving $c_i \|w_i\|^2 = 0$, so $c_i = 0$.

(b) Write $v = \sum_i c_i e_i$ and take the inner product with e_j: $\langle e_j, v\rangle = \sum_i c_i \langle e_j, e_i\rangle = c_j$.

(c) Substitute (b) for v and w and expand; only the terms with matching indices survive (*Ex*). □

5.3 Orthogonal complements

For a subset $S \subseteq V$, its *orthogonal complement* is $S^\perp = \{v \in V : \langle v, s\rangle = 0 \text{ for all } s \in S\}$, the vectors perpendicular to everything in S. It is a subspace (a sum or multiple of vectors perpendicular to s is perpendicular to s), even when S is not. For example, in $\mathbb{R}^3$ the complement of a line through 0 is the perpendicular plane, and vice versa.

Definition 5.3

Theorem 5.2 For every subspace W of a Euclidean space V: $V = W \oplus W^\perp$, and $(W^\perp)^\perp = W$.

Proof 1 (class, Oct 15): dimension count. Let $\dim V = n$ and $\dim W = k$.

The intersection is 0. A vector $w \in W \cap W^\perp$ is perpendicular to itself, so $\langle w, w\rangle = 0$ and $w = 0$.

$\dim W^\perp \geq n - k$. Let $w_1, \dots, w_k$ be a basis of W. A vector is perpendicular to all of W as soon as it is perpendicular to each w_i, so

$$W^\perp = \ker\langle w_1, -\rangle \cap \cdots \cap \ker\langle w_k, -\rangle,$$

where $\langle w_i, -\rangle : V \to \mathbb{R}$ is the linear functional $x \mapsto \langle w_i, x\rangle$. A linear functional has rank at most 1, so by rank–nullity its kernel has codimension at most 1 in any subspace it is restricted to. Hence $N_1 = \ker\langle w_1, -\rangle$ has dimension $\geq n-1$, $N_2 = N_1 \cap \ker\langle w_2, -\rangle$ has dimension $\geq n-2$, and so on; $N_k = W^\perp$ has dimension $\geq n - k$.

Conclusion.

$$n \geq \dim(W + W^\perp) = \dim W + \dim W^\perp - \dim(W \cap W^\perp) \geq k + (n-k) - 0 = n,$$

so equality holds throughout: $W + W^\perp = V$, and the sum is direct.

$(W^\perp)^\perp = W$. Every $w \in W$ is perpendicular to every vector of $W^\perp$, so $W \subseteq (W^\perp)^\perp$. Applying the first part to $W^\perp$ gives $\dim(W^\perp)^\perp = n - \dim W^\perp = n - (n-k) = k$, so the inclusion is an equality. □

Proof 2 (Oct 22): linear equations. Fix a basis of V. The conditions $\langle w_i, x\rangle = 0$ $(i = 1, \dots, k)$ are k homogeneous linear equations in the n coordinates of x, so their solution space $W^\perp$ has dimension at least $n - k$ (rank–nullity for the $k \times n$ coefficient matrix). Continue as in Proof 1. □

Proof 3 (Gram–Schmidt; Axler 6.53): an explicit formula. Let $e_1, \dots, e_k$ be an ON basis of W (Gram–Schmidt, section 5.5, applied to a basis of W). For $v \in V$, put

$$u = \sum_{i=1}^{k} \langle e_i, v\rangle\, e_i \in W, \qquad w = v - u.$$

Then for each j, $\langle e_j, w\rangle = \langle e_j, v\rangle - \sum_i \langle e_i, v\rangle\langle e_j, e_i\rangle = \langle e_j, v\rangle - \langle e_j, v\rangle = 0$, so w is perpendicular to a basis of W, hence $w \in W^\perp$. So $v = u + w$ with $u \in W$, $w \in W^\perp$, and with $W \cap W^\perp = 0$ this gives $V = W \oplus W^\perp$. This proof also gives the formula for P_W below. □

Another proof of $(W^\perp)^\perp = W$ without dimensions. Let $v \in (W^\perp)^\perp$ and write $v = u + w$ with $u \in W$, $w \in W^\perp$. Since $u \in W \subseteq (W^\perp)^\perp$, also $w = v - u \in (W^\perp)^\perp$. So w lies in both $W^\perp$ and $(W^\perp)^\perp$, hence is perpendicular to itself, so $w = 0$ and $v = u \in W$. □

Corollary 5.3.

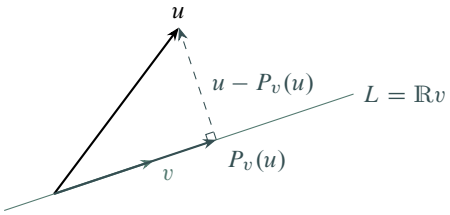
1. *There are many ON bases. Take $e_1 \neq 0$, then $e_2 \in e_1^\perp$ (of dimension $n-1$), then $e_3 \in \{e_1, e_2\}^\perp$ (of dimension $n-2$), and so on, and scale every e_i to norm 1. Any ON list extends to an ON basis.*
2. *Given a subspace W and $v \in V$, write $v = w_1 + w_2$ uniquely with $w_1 \in W$ and $w_2 \in W^\perp$. The map $P_W(v) = w_1$ is the orthogonal projection onto W. It is linear, $P_W^2 = P_W$, $\operatorname{im} P_W = W$, $\ker P_W = W^\perp$, $\|P_W v\| \leq \|v\|$, and $P_{W^\perp} = I - P_W$. If $e_1, \dots, e_k$ is an ON basis of W, then $P_W(v) = \sum_i \langle e_i, v\rangle\, e_i$ (Ex).*

5.4 Projection onto a line, Cauchy–Schwarz

Proposition 5.4 (Projection onto a line). *For a fixed $v \neq 0$, the orthogonal projection onto $L = \mathbb{R}v$ is the linear map $V \to L$,*

$$P_v(u) = \frac{\langle v, u\rangle}{\|v\|^2}\, v.$$

So $u = P_v(u) + \big(u - P_v(u)\big)$, the component of u along v plus the component of u perpendicular to v.



Proof. We want $u = \lambda v + (u - \lambda v)$ with the second part perpendicular to v. The condition $\langle v, u - \lambda v\rangle = 0$ reads $\langle v, u\rangle - \lambda\|v\|^2 = 0$, so $\lambda = \langle v, u\rangle / \|v\|^2$, and this λ is the only choice. □

Example 5.2

In $\mathbb{R}^3$, project $u = (1, 2, 3)$ onto $v = (1, 1, 1)$. Then $\langle v, u\rangle = 6$ and $\|v\|^2 = 3$, so $P_v(u) = (2, 2, 2)$, and $u - P_v(u) = (-1, 0, 1)$ is perpendicular to v.

Theorem 5.5

For $u, v \in V$:

1. (Pythagoras) If $v \perp w$, then $\|v + w\|^2 = \|v\|^2 + \|w\|^2$.
2. (Cauchy–Schwarz) $|\langle v, u\rangle| \leq \|v\|\|u\|$. For $v \neq 0$, equality holds if and only if $u \in \mathbb{R}v$.

Theorem 5.5 *continued*

3. (Triangle inequality) $\|u+v\| \le \|u\| + \|v\|$, with equality if and only if one of u, v is a nonnegative multiple of the other.
4. (Parallelogram law) $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$.

Proof of (b), 1 (class, Oct 20): projection and Pythagoras. Apply Pythagoras to $u = P_v(u) + (u - P_v(u))$. This gives

$$\|u\|^2 \ge \left\| \frac{\langle v,u\rangle}{\|v\|^2} v \right\|^2 = \frac{\langle v,u\rangle^2}{\|v\|^2},$$

with equality if and only if $u - P_v(u) = 0$. □

Proof of (b), 2 (Oct 20): "bash it out" in an ON basis. Let e_1, e_2 be an ON basis of a plane containing u and v (or work in an ON basis of V). By Parseval, $\langle v,u\rangle = \sum v_i u_i$, so Cauchy–Schwarz becomes $(\sum v_i u_i)^2 \le (\sum v_i^2)(\sum u_i^2)$. This follows from Lagrange's identity

$$\Big(\sum v_i^2\Big)\Big(\sum u_i^2\Big) - \Big(\sum v_i u_i\Big)^2 = \sum_{i<j} (v_i u_j - v_j u_i)^2 \ (Ex).$$

□

Proof of (b), 3: a quadratic in t. For real t, $0 \le \|u - tv\|^2 = \|u\|^2 - 2t\langle v,u\rangle + t^2\|v\|^2$. A quadratic in t that is never negative has discriminant ≤ 0: $4\langle v,u\rangle^2 - 4\|u\|^2\|v\|^2 \le 0$. Equality means the quadratic has a real root t_0, so $u = t_0 v$ (Ex). □

Proof of (b), 4: normalize. Both sides scale the same way, so assume $\|u\| = \|v\| = 1$. Then $0 \le \|u \mp v\|^2 = 2 \mp 2\langle u,v\rangle$ gives $|\langle u,v\rangle| \le 1$, with equality iff $u = \pm v$ (Ex). □

Proof of (a), (c), (d). (a) $\|v+w\|^2 = \|v\|^2 + 2\langle v,w\rangle + \|w\|^2$, and $\langle v,w\rangle = 0$.

(c) $\|u+v\|^2 = \|u\|^2 + 2\langle u,v\rangle + \|v\|^2 \le \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$, using $\langle u,v\rangle \le |\langle u,v\rangle| \le \|u\|\|v\|$. Equality needs $\langle u,v\rangle = \|u\|\|v\|$: by the equality case of Cauchy–Schwarz $u = tv$ (if $v \ne 0$), and then $t\|v\|^2 = |t|\|v\|^2$ forces $t \ge 0$.

(d) Add the expansions of $\|u+v\|^2$ and $\|u-v\|^2$; the cross terms $\pm 2\langle u,v\rangle$ cancel. □

5.5 The Gram–Schmidt process

Gram–Schmidt — Theorem 5.6

Theorem 5.6 *continued*

Let $v_1, \dots, v_n$ be a basis. Set $e_1 = v_1/\|v_1\|$, and for $k \geq 1$

$$e_{k+1} = \frac{v_{k+1} - \sum_{i=1}^{k} \langle e_i, v_{k+1}\rangle\, e_i}{\|\text{numerator}\|}.$$

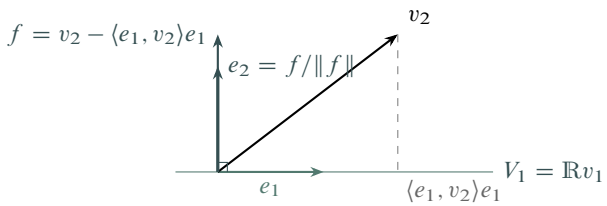
Then $e_1, \dots, e_n$ is an ON basis, and $\operatorname{span}\{v_1, \dots, v_k\} = \operatorname{span}\{e_1, \dots, e_k\}$ for $k = 1, \dots, \dim V$. So the change-of-basis matrix, in either direction, is upper triangular.

Proof. Induction on k: assume $e_1, \dots, e_k$ are orthonormal with $\operatorname{span}(e_1, \dots, e_k) = \operatorname{span}(v_1, \dots, v_k) =: V_k$. Let $f = v_{k+1} - \sum_{i \leq k} \langle e_i, v_{k+1}\rangle e_i$ be the numerator.

1. $f \neq 0$: otherwise $v_{k+1} \in V_k = \operatorname{span}(v_1, \dots, v_k)$, contradicting independence.
2. $f \perp e_j$ *for* $j \leq k$: $\langle e_j, f\rangle = \langle e_j, v_{k+1}\rangle - \langle e_j, v_{k+1}\rangle = 0$, as in Proof 3 of theorem 5.2. So $e_{k+1} = f/\|f\|$ is a unit vector perpendicular to $e_1, \dots, e_k$.
3. *Spans:* e_{k+1} is a combination of v_{k+1} and $e_1, \dots, e_k \in V_k$, so lies in V_{k+1}; conversely $v_{k+1} = \|f\| e_{k+1} + \sum_{i \leq k} \langle e_i, v_{k+1}\rangle e_i$ lies in $\operatorname{span}(e_1, \dots, e_{k+1})$.

Since $v_k \in \operatorname{span}(e_1, \dots, e_k)$ for each k, the matrix expressing the v's in terms of the e's is upper triangular, and so is its inverse. In terms of orthogonal projection (theorem 5.2 and its corollary), $f = v_{k+1} - P_{V_k}(v_{k+1})$ is the component of v_{k+1} perpendicular to V_k. □

One step of the process, for $k = 1$: subtract from v_2 its component along e_1, then rescale.



Example 5.3

In $\mathbb{R}^3$, take $v_1 = (1, 0, 1)$ and $v_2 = (1, 1, 0)$. Then $e_1 = \frac{1}{\sqrt{2}}(1, 0, 1)$ and $\langle e_1, v_2\rangle = \frac{1}{\sqrt{2}}$. So the numerator is $v_2 - \frac{1}{2}(1, 0, 1) = (\frac{1}{2}, 1, -\frac{1}{2})$, and $e_2 = \frac{1}{\sqrt{6}}(1, 2, -1)$.

Corollary 5.7.

1. *If an operator T on an inner product space has an upper triangular matrix M in some basis, then it has an upper triangular matrix in an ON basis, namely $P^{-1}MP$, where P is the (upper triangular) change of basis given by Gram–Schmidt. In particular, every complex matrix is unitarily triangularizable (Schur).*

2. *Gram–Schmidt shows $V = W \oplus W^\perp$ explicitly: extend a basis of W to a basis of V and apply Gram–Schmidt (Ex).*

Another proof of (a). A basis gives an upper triangular matrix exactly when the subspaces $V_k = \operatorname{span}(v_1, \dots, v_k)$ are T-invariant (Axler 5C.10). Gram–Schmidt does not change the V_k. □

Exercises

From the board and the tutorials

5.5.1 *Oct 14 board* **[3]** Show that the angle $\theta \in [0, \pi]$ is well defined (by Cauchy–Schwarz), and that it agrees with the geometric angle in $\mathbb{R}^2$ and $\mathbb{R}^3$.

5.5.2 *Oct 15 board* **[1]** Find an ON basis for the polynomials of degree at most 2 with $\langle f, g\rangle = \int_{-1}^{1} fg$, by Gram–Schmidt on $1, x, x^2$ (Artin 8.4.9). What about degree at most n?

5.5.3 *Oct 15, 22 boards* **[1]** Prove $\dim W^\perp \geq n - \dim W$ in both ways, and $(W^\perp)^\perp = W$.

5.5.4 *Oct 15, 20 boards* **[1]** Give two proofs of Cauchy–Schwarz: one reducing to the dot product, one using the projection formula. Find the equality cases of Cauchy–Schwarz and of the triangle inequality. Prove the parallelogram law.

5.5.5 *Oct 15 board* **[2]** Fix $v \neq 0$ and find the formula for the projection onto $\mathbb{R}v$ by writing $u =$ (component along v) + (component perpendicular to v).

From the books

5.5.6 *Artin 8.4.1, 8.4.3* **[2]** (a) What is the inverse of a matrix whose columns are orthogonal (and nonzero)? (b) For a real $m \times n$ matrix A, prove that $B = A^T A$ satisfies $X^T B X \geq 0$ for all X, and that $\operatorname{rank} A = \operatorname{rank} B$.

5.5.7 *Artin 8.4.6, 8.4.7* **[1]** (a) Extend $X_1 = \frac{1}{2}(1, -1, 1, 1)^T$ to an ON basis of $\mathbb{R}^4$. (b) Apply Gram–Schmidt to $(1, 1, 0)^T, (1, 0, 1)^T, (0, 1, 1)^T$.

5.5.8 *Artin 8.4.10, 8.4.11* **[2]** (a) On real $n \times n$ matrices, show that $\langle A, B\rangle = \operatorname{tr}(A^T B)$ is an inner product, and find an ON basis. (b) For subspaces W_1, W_2, prove $(W_1 + W_2)^\perp = W_1^\perp \cap W_2^\perp$, $W \subseteq W^{\perp\perp}$, and $W_1 \subseteq W_2 \Rightarrow W_1^\perp \supseteq W_2^\perp$. Also prove $(W_1 \cap W_2)^\perp = W_1^\perp + W_2^\perp$ in a Euclidean space.

5.5.9 *Artin 8.4.15, 8.4.16* **[1]** (a) Let $W = \operatorname{span}\{(1,1,0)^T, (0,1,1)^T\} \subseteq \mathbb{R}^3$. Find the orthogonal projection of $(1,0,0)^T$ to W. (b) With $\langle A, B\rangle = \operatorname{tr}(A^T B)$ on real 3×3 matrices and W the skew-symmetric matrices, project $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 1 & 3 & 0 \end{bmatrix}$ to W. (What is $W^\perp$?)

5.5.10 *Artin 8.4.19* **[3]** Let W be a plane in $\mathbb{R}^3$, π the orthogonal projection onto W, and $(a_i, b_i)^T$ the coordinates of $\pi(e_i)$ in an ON basis of W. Prove that (a_1, a_2, a_3) and (b_1, b_2, b_3) are orthogonal unit vectors.

5.5.11 *Artin 8.5.1, 8.5.2* **[2]** (a) Prove that if $\|v\| = \|w\|$, then $v + w \perp v - w$. (b) Prove $W = W^{\perp\perp}$ for a subspace of a Euclidean space.

5.5.12 *Axler 6A.1, 6A.2, 6A.3* **[3]** (a) Prove or disprove: $\sum_{j,k}\langle v_j, v_k\rangle \geq 0$ for any $v_1, \ldots, v_m$. (b) For an operator S, show that $\langle u, v\rangle_1 = \langle Su, Sv\rangle$ is an inner product if and only if S is injective. (c) Show that $|x_1 y_1| + |x_2 y_2|$ is not an inner product on $\mathbb{R}^2$, and $x_1 y_1 + x_3 y_3$ is not one on $\mathbb{R}^3$.

5.5.13 *Axler 6A.6, 6A.7* **[3]** (a) Prove $\langle u, v\rangle = 0 \iff \|u\| \leq \|u + av\|$ for all scalars a. (b) Prove $\|au + bv\| = \|bu + av\|$ for all real a, b if and only if $\|u\| = \|v\|$.

5.5.14 *Axler 6A.8, 6A.12, 6A.13* **[2]** (a) If $a^2 + b^2 + c^2 + x^2 + y^2 \leq 1$, prove $a+b+c+4x+9y \leq 10$. (b) For positive a, b, c, d, prove $(a+b+c+d)(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}) \geq 16$, and find when equality holds. (c) Show that the square of the average of $a_1, \ldots, a_n$ is at most the average of the squares.

5.5.15 *Axler 6A.10, 6A.17* **[3]** (a) If $\|u\|, \|v\| \leq 1$, prove $\sqrt{1-\|u\|^2}\sqrt{1-\|v\|^2} \leq 1 - |\langle u, v\rangle|$. (b) Prove $\left(\sum a_k b_k\right)^2 \leq \left(\sum k a_k^2\right)\left(\sum b_k^2/k\right)$ for real a_k, b_k.

5.5.16 *Axler 6A.20, 6A.21, 6A.22* **[3]** (a) Prove $\big|\|u\| - \|v\|\big| \leq \|u - v\|$. (b) If $\|u\| = 3$, $\|u + v\| = 4$ and $\|u - v\| = 6$, find $\|v\|$. (c) Prove $\|u + v\|\,\|u - v\| \leq \|u\|^2 + \|v\|^2$.

5.5.17 *Axler 6A.31, 6A.32* **[3]** (a) Prove $\|w - \frac{1}{2}(u + v)\|^2 = \frac{\|w-u\|^2 + \|w-v\|^2}{2} - \frac{\|u-v\|^2}{4}$ (Apollonius). (b) Let $E \subseteq V$ be closed under midpoints ($u, v \in E \Rightarrow \frac{1}{2}(u + v) \in E$). Show that for each w there is at most one point of E closest to w.

5.5.18 *Axler 6B.1, 6B.3* **[3]** (a) If $\|a_1 e_1 + \cdots + a_m e_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$ for all scalars, show that $e_1, \ldots, e_m$ is orthonormal. (b) For an orthonormal list $e_1, \ldots, e_m$, prove $\|v\|^2 = \sum |\langle v, e_k\rangle|^2$ if and only if $v \in \operatorname{span}(e_1, \ldots, e_m)$ (Bessel with equality).

5.5.19 *Axler 6B.6* **[4]** Let $e_1, \ldots, e_n$ be an ON basis. (a) If $\|e_k - v_k\| < \frac{1}{\sqrt{n}}$ for each k, prove $v_1, \ldots, v_n$ is a basis. (b) Show that $\frac{1}{\sqrt{n}}$ cannot be increased.

5.5.20 *Axler 6B.7, 6B.8* **[2]** (a) T on $\mathbb{R}^3$ is upper triangular in the basis $(1,0,0)$, $(1,1,1)$, $(1,1,2)$. Find an ON basis in which T is upper triangular. (b) On polynomials of degree ≤ 2 with $\langle p,q\rangle = \int_0^1 pq$, apply Gram–Schmidt to $1, x, x^2$, and find the matrix of differentiation in the resulting basis; check that it is upper triangular.

5.5.21 *Axler 6B.9, 6B.10, 6B.13* **[3]** (a) If $e_1,\dots,e_m$ come from Gram–Schmidt on $v_1,\dots,v_m$, prove $\langle v_k, e_k\rangle > 0$. (b) Show that Gram–Schmidt gives the only ON list with $\langle v_k, e_k\rangle > 0$ and $\operatorname{span}(v_1,\dots,v_k) = \operatorname{span}(e_1,\dots,e_k)$ for all k. (c) Show that $v_1,\dots,v_m$ is dependent if and only if Gram–Schmidt produces a zero numerator at some step.

5.5.22 *Axler 6B.14* **[3]** For an independent list $v_1,\dots,v_m$ in a real inner product space, prove there are exactly 2^m ON lists $e_1,\dots,e_m$ with $\operatorname{span}(v_1,\dots,v_k) = \operatorname{span}(e_1,\dots,e_k)$ for all k.

5.5.23 *Axler 6B.18, 6B.19* **[3]** (a) If $u_1,\dots,u_m$ is independent, show there is v with $\langle u_k, v\rangle = 1$ for all k. (b) If $v_1,\dots,v_n$ is a basis, show there is a basis $u_1,\dots,u_n$ with $\langle v_j, u_k\rangle = \delta_{jk}$ (the dual basis inside V).

5.5.24 *Axler 6C.1, 6C.2, 6C.3* **[2]** (a) Prove $\{v_1,\dots,v_m\}^\perp = (\operatorname{span}(v_1,\dots,v_m))^\perp$. (b) If Gram–Schmidt is applied to a basis $u_1,\dots,u_m,v_1,\dots,v_n$ where the u's span U, show the first m outputs form an ON basis of U and the rest an ON basis of $U^\perp$. (c) Find ON bases of $U = \operatorname{span}((1,2,3,-4),(-5,4,3,2)) \subseteq \mathbb{R}^4$ and of $U^\perp$.

5.5.25 *Axler 6C.4, 6C.5* **[3]** (a) If $\|e_k\| = 1$ for each k and $\|v\|^2 = \sum_k |\langle v, e_k\rangle|^2$ for all v, prove $e_1,\dots,e_n$ is an ON basis. (b) Prove $P_{U^\perp} = I - P_U$.

5.5.26 *Axler 6C.7, 6C.9, 6C.10* **[2]** (a) Prove $P_X P_Y = 0$ if and only if $X \perp Y$. (b) If $P^2 = P$ and every vector of $\ker P$ is orthogonal to every vector of $\operatorname{im} P$, prove $P = P_U$ for some U. (c) If $P^2 = P$ and $\|Pv\| \le \|v\|$ for all v, prove $P = P_U$ for some U.

5.5.27 *Axler 6C.11, 6C.12* **[3]** (a) Prove U is T-invariant if and only if $P_U T P_U = T P_U$. (b) Prove U and $U^\perp$ are both T-invariant if and only if $P_U T = T P_U$.

5.5.28 *Axler 6C.15* **[2]** In $\mathbb{R}^4$ with $U = \operatorname{span}((1,1,0,0),(1,1,1,2))$, find $u \in U$ minimizing $\|u - (1,2,3,4)\|$. (The minimizer is P_U of the vector.)

Harder or unusual

5.5.29 *Napkin 13C* **[4]** (Oddtown.) A town has n people and k clubs. Every club has an odd number of members, and every two clubs share an even number of members. Prove $k \le n$. (Hint: vectors in $\mathbb{F}_2^n$ and the dot product.)

5.5.30 *Artin 8.M.14* **[4]** What is the largest number of vectors in $\mathbb{R}^n$ with $v_i \cdot v_j < 0$ for all $i \neq j$?

5.5.31 *Axler 6A.28, Halmos §64.6* **[4]** Prove that a norm satisfying the parallelogram law comes from an inner product. (Define $\langle u, v\rangle$ by polarization; additivity is the hard part.)

5.5.32 *Napkin 13G, Axler 6A.25, Halmos §64.5* **[3]** Show that $\|(x_1, \ldots, x_n)\| = \sum |x_i|$ ($n \geq 2$) and $\max |x_i|$ are norms on $\mathbb{R}^n$ that do not come from inner products. For $p > 0$, show $(|x|^p + |y|^p)^{1/p}$ comes from an inner product on $\mathbb{R}^2$ if and only if $p = 2$.

5.5.33 *Halmos §64.3, §64.4* **[3]** (a) If $|\langle x, y\rangle| = \|x\|\|y\|$, prove x, y are dependent. (b) Suppose $\langle \cdot, \cdot \rangle$ is bilinear and symmetric with $\langle x, x\rangle \geq 0$ but possibly $= 0$ for $x \neq 0$. Prove Cauchy–Schwarz still holds, that $\{x : \langle x, x\rangle = 0\}$ is a subspace N, and that $\langle \cdot, \cdot \rangle$ induces an honest inner product on V/N.

5.5.34 *Axler 6B.15, 6B.16* **[4]** (a) If two inner products on V have the same pairs of orthogonal vectors, prove one is a positive multiple of the other. (b) For two inner products on a finite-dimensional V, prove $\|v\|_1 \leq c\|v\|_2$ for some $c > 0$ and all v.

5.5.35 *Axler 6A.23* **[4]** If $\|v_k\| \leq 1$ for $k = 1, \ldots, m$, prove there are signs $a_k \in \{1, -1\}$ with $\|a_1 v_1 + \cdots + a_m v_m\| \leq \sqrt{m}$.

5.5.36 *Halmos §62.3* **[4]** On polynomials of degree $\leq n$ with $\langle x, y\rangle = \int_0^1 x(t)y(t)\,dt$, find a polynomial of degree n orthogonal to $1, t, \ldots, t^{n-1}$. (Try $\frac{d^n}{dt^n}\left(t^n(1-t)^n\right)$ and integrate by parts.)

6 Hermitian Spaces

6.1 Complex inner products

Definition 6.1

An *inner product space* V over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ has an inner product $\langle \cdot, \cdot \rangle \colon V \times V \to \mathbb{F}$ satisfying

- $\langle v, v \rangle$ is real and ≥ 0, with $\langle v, v \rangle = 0 \iff v = 0$;
- $\langle v, w \rangle = \overline{\langle w, v \rangle}$ (complex conjugate);
- $\langle cv_1 + v_2, w \rangle = \bar{c}\langle v_1, w \rangle + \langle v_2, w \rangle$ (conjugate-linear in the *first* slot);
- $\langle v, cw_1 + w_2 \rangle = c\langle v, w_1 \rangle + \langle v, w_2 \rangle$ (linear in the *second* slot).

This is the *opposite* of LADR's convention, so rewrite Axler's formulas before using them. Over $\mathbb{R}$ it is the definition of the previous chapter. A finite-dimensional complex inner product space is a *Hermitian space*.

Example 6.1

On $\mathbb{F}^n$, $\langle v, w \rangle = \bar{v}^T w = \sum_i \bar{v}_i w_i$ (*Ex*). On $M_n(\mathbb{C})$, $\langle A, B \rangle = \operatorname{tr}(A^* B) = \sum_{i,j} \bar{a}_{ij} b_{ij}$, where $A^* = \bar{A}^T$. Any subspace of an inner product space is again one.

Remark 6.1.1.

1. Perpendicularity is symmetric: $\langle v, w \rangle = 0 \iff \langle w, v \rangle = 0$. For $S \subseteq V$, $S^\perp = \{v : v \perp s \ \forall s \in S\}$ is a subspace.
2. $\|v\| = \sqrt{\langle v, v \rangle}$ is real, and positive unless $v = 0$; and $\|cv\| = |c|\|v\|$.
3. In an ON basis $e_1, \ldots, e_n$, write $[v]$ for the coordinate vector of v. Then $\langle v, w \rangle = \overline{[v]}^T [w] = [v]^* [w]$.

6.2 What carries over

Everything in the previous chapter carries over, provided the vectors are written in the order shown:

- $V = W \oplus W^\perp$ and $(W^\perp)^\perp = W$. $W \cap W^\perp = 0$ by positive definiteness, and the dimension count uses that $\langle w_i, -\rangle$ is linear.
- Any set of mutually perpendicular nonzero vectors is independent. In an ON basis, $v = \sum_i \langle e_i, v\rangle\, e_i = \sum_i \overline{\langle v, e_i\rangle}\, e_i$.
- Any ON list can be completed to an ON basis, using $V = W \oplus W^\perp$ or Gram–Schmidt. Gram–Schmidt keeps the same formula.
- Projection along a nonzero v: $P_v(u) = \frac{\langle v,u\rangle}{\|v\|^2} v$, and $u - P_v(u) \perp v$.
- Pythagoras; Cauchy–Schwarz $|\langle v, u\rangle| \le \|u\|\|v\|$, with equality iff $u \in \mathbb{F}v$ (assuming $v \neq 0$); the triangle inequality, with equality iff $u \in \mathbb{R}_{\ge 0} v$.

Remark 6.2.1. In the projection, $\langle v, u - \lambda v\rangle = \langle v, u\rangle - \lambda\|v\|^2$. If v were in the second slot, this would give $\bar\lambda$ instead. Find where each proof uses the order of the slots (*Ex*).

Proposition 6.1 (Polarization). *Over* $\mathbb{R}$: $\langle v, w\rangle = \frac{1}{4}\big(\|v + w\|^2 - \|v - w\|^2\big)$. *Over* $\mathbb{C}$, *with the course convention,*

$$\langle v, w\rangle = \frac{1}{4}\sum_{k=0}^{3} i^{-k}\|v + i^k w\|^2 = \frac{1}{4}\Big(\|v+w\|^2 - \|v-w\|^2 - i\,\|v+iw\|^2 + i\,\|v-iw\|^2\Big).$$

So the inner product is determined by the norm.

Proof outline. $\|v + cw\|^2 = \|v\|^2 + |c|^2\|w\|^2 + c\langle v, w\rangle + \bar c\,\overline{\langle v, w\rangle}$. Multiply by i^{-k} with $c = i^k$ and sum over k; use $\sum_{k=0}^{3} i^{-k} = \sum_k i^{-2k} = 0$ (*Ex*). (Axler 6A.27 has the signs of the i-terms reversed, because of the other convention.) □

6.3 Other proofs

Cauchy–Schwarz over $\mathbb{C}$ *by rotating.* Choose θ with $e^{i\theta}\langle v, u\rangle \ge 0$ and replace u by $e^{i\theta}u$; then $\langle v, u\rangle$ is real and the real proofs (the quadratic in t, or normalizing) apply, since $\|u - tv\|^2 = \|u\|^2 - 2t\,\mathrm{Re}\langle v, u\rangle + t^2\|v\|^2$ for real t (*Ex*).

The real part. For a complex inner product, $\mathrm{Re}\langle v, w\rangle$ is a real inner product on V viewed as a real vector space of dimension $2\dim_{\mathbb{C}} V$, with the same norm, and $\langle v, w\rangle = \mathrm{Re}\langle v, w\rangle + i\,\mathrm{Re}\langle iv, w\rangle$ (*Ex*). This turns many complex statements into real ones.

Exercises

From the board and the tutorials

6.3.1 *Oct 22 board* **[1]** Check the axioms for $\langle v, w\rangle = \bar{v}^T w$ on $\mathbb{C}^n$, and prove $\langle v, w\rangle = [v]^*[w]$ in an ON basis.

6.3.2 *Oct 22 board* **[1]** Redo Cauchy–Schwarz, its equality case and the triangle inequality over $\mathbb{C}$, with the course convention. Check that Gram–Schmidt "stays the same".

6.3.3 *Oct 22 board* **[2]** Prove $\dim W^\perp \geq n - \dim W$ over $\mathbb{C}$ by viewing $W^\perp$ as a solution space of homogeneous linear equations. Where does the order of the slots matter?

From the books

6.3.4 *Axler 6A.27, Halmos §62.2* **[2]** Prove proposition 6.1, and rewrite Axler's version $\langle u, v\rangle = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2)$ in the course convention.

6.3.5 *Halmos §62.4* **[3]** (a) In a real inner product space, $x \perp y$ if and only if $\|x+y\|^2 = \|x\|^2 + \|y\|^2$. (b) Show (a) fails over $\mathbb{C}$. (c) In a complex inner product space, $x \perp y$ if and only if $\|\alpha x + \beta y\|^2 = \|\alpha x\|^2 + \|\beta y\|^2$ for all scalars α, β. (d) If $\|x\| = \|y\|$ in a real space, then $x + y \perp x - y$. What is true over $\mathbb{C}$?

6.3.6 *Halmos §62.1* **[3]** For which complex numbers $\alpha, \beta, \gamma, \delta$ does $\langle x, y\rangle = \alpha\bar{x}_1 y_1 + \beta\bar{x}_2 y_1 + \gamma\bar{x}_1 y_2 + \delta\bar{x}_2 y_2$ define an inner product on $\mathbb{C}^2$?

6.3.7 *Artin 8.4.13(a), Axler 6A.29* **[1]** (a) Show that $\langle A, B\rangle = \operatorname{tr}(A^*B)$ is an inner product on complex $n\times n$ matrices, and find an ON basis (2025 Test 2 Q5). (b) If $V_1, \dots, V_m$ are inner product spaces, show $\langle (u_k), (v_k)\rangle = \sum_k \langle u_k, v_k\rangle$ is an inner product on $V_1 \times \dots \times V_m$.

6.3.8 *Axler 6A.30* **[4]** Let V be a real inner product space and $V_{\mathbb{C}} = \{u + iv : u, v \in V\}$ its complexification. Find the formula for an inner product on $V_{\mathbb{C}}$ extending that of V, in the course convention, with $\|u+iv\|^2 = \|u\|^2 + \|v\|^2$, and verify the axioms.

7 Orthogonal and Unitary Operators

7.1 Definition and basic properties

Definition 7.1

An operator T on an inner product space is *orthogonal* ($\mathbb{F} = \mathbb{R}$) or *unitary* ($\mathbb{F} = \mathbb{C}$) if it preserves the inner product:

$$\langle Tv, Tw\rangle = \langle v, w\rangle \quad \text{for all } v, w \in V.$$

A square matrix A is orthogonal if $A^TA = I$ (real) and unitary if $A^*A = I$ (complex).

Proposition 7.1. *Let T be orthogonal or unitary, and let A be its matrix with respect to an ON basis $e_1, \dots, e_n$.*

1. *T preserves perpendicularity and norm. In particular $Tv = 0 \Rightarrow v = 0$, so T is injective and hence an isomorphism (on a finite-dimensional space).*
2. *Conversely, a linear T with $\|Tv\| = \|v\|$ for all v is orthogonal or unitary.*
3. *T is orthogonal $\iff A^TA = I \iff A^{-1} = A^T \iff AA^T = I \iff$ the columns of A are orthonormal for the dot product $\iff$ the rows of A are. T is unitary $\iff A^*A = I$.*
4. *If T is orthogonal, $\det A = \pm 1$, since $A^TA = I$ gives $\det(A^T)\det(A) = \det(A)^2 = 1$. If T is unitary, $|\det A| = 1$.*
5. *Every eigenvalue λ of T has $|\lambda| = 1$.*
6. *If W is T-invariant, then so is $W^\perp$.*

Proof (class). (a) $\|Tv\|^2 = \langle Tv, Tv\rangle = \langle v, v\rangle$, and $\langle Tv, Tw\rangle = 0$ whenever $\langle v, w\rangle = 0$. If $Tv = 0$ then $\|v\| = \|Tv\| = 0$; an injective operator on a finite-dimensional space is bijective.

(b) Polarization (proposition 6.1) writes $\langle v, w\rangle$ using norms only, and T preserves norms and is linear, so $\langle Tv, Tw\rangle$ is given by the same expression with Tv, Tw, which has the same value.

(c) In an ON basis $\langle v, w\rangle = [v]^*[w]$ and $[Tv] = A[v]$, so $\langle Tv, Tw\rangle = [v]^* A^* A[w]$. Taking $v = e_i$, $w = e_j$ shows that $\langle Tv, Tw\rangle = \langle v, w\rangle$ for all v, w exactly when $(A^*A)_{ij} = \delta_{ij}$, that is, $A^*A = I$. The (i, j) entry of A^*A is the inner product of column i with column j, so this says the columns are orthonormal. For square matrices $A^*A = I$ means $A^{-1} = A^*$, which is equivalent to $AA^* = I$, and that says the rows are orthonormal. Over $\mathbb{R}$, $A^* = A^T$.

(e) If $Tv = \lambda v$ with $v \neq 0$, then $\langle v, v\rangle = \langle Tv, Tv\rangle = \langle \lambda v, \lambda v\rangle = \bar{\lambda}\lambda\langle v, v\rangle$, so $|\lambda| = 1$. For a real orthogonal matrix, the complex eigenvalues also satisfy $|\lambda| = 1$, since A is unitary on $\mathbb{C}^n$.

(f) We must show $Tu \perp W$ for every $u \in W^\perp$. Since T is injective and $T(W) \subseteq W$, the restriction $T|_W$ is injective, hence onto W: every $w \in W$ is Tw' for some $w' \in W$. Then $\langle Tu, w\rangle = \langle Tu, Tw'\rangle = \langle u, w'\rangle = 0$. □

Another proof of (b). Expand $\|T(v + w)\|^2 = \|v + w\|^2$: this gives $\mathrm{Re}\langle Tv, Tw\rangle = \mathrm{Re}\langle v, w\rangle$. Over $\mathbb{C}$, replace w by iw to get the imaginary parts (*Ex*). □

Another proof of (f). T^{-1} is also orthogonal/unitary, and $\langle Tu, w\rangle = \langle T^{-1}Tu, T^{-1}w\rangle = \langle u, T^{-1}w\rangle$. Since $T^{-1}(W) = W$, this is 0 for $u \in W^\perp$, $w \in W$. □

Another proof of (e) for real orthogonal matrices. If $AX = \lambda X$ with $X \in \mathbb{C}^n$ nonzero, take conjugates: $A\bar{X} = \bar{\lambda}\bar{X}$. Then $\bar{X}^T X = \bar{X}^T A^T AX = (A\bar{X})^T(AX) = \bar{\lambda}\lambda\, \bar{X}^T X$, and $\bar{X}^T X = \sum |x_i|^2 > 0$ (*Ex*). (Compare Artin 8.M.6, where $X^T X$ is used instead and the argument fails.) □

7.2 The orthogonal groups

Definition 7.2

O_n, the *orthogonal group*, is the set of all orthogonal $n \times n$ matrices (or operators). If $A, B \in O_n$, then $AB \in O_n$ and $A^{-1} = A^T \in O_n$. The *special orthogonal group* $SO_n = \{A \in O_n : \det A = 1\}$ is a subgroup, since it is closed under multiplication and inverses. Similarly U_n is the group of unitary matrices.

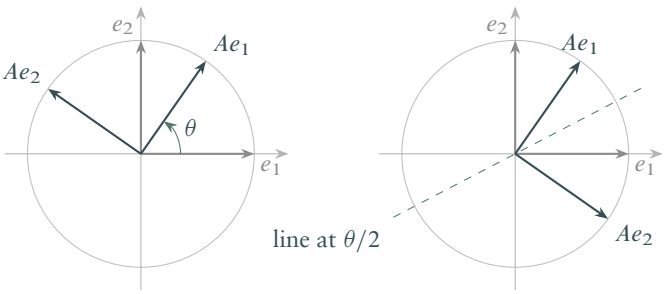
Theorem 7.2

Oct 21

1. $O_n = SO_n \sqcup M \cdot SO_n$, where M is any fixed orthogonal matrix with $\det M = -1$, for example $M = \mathrm{diag}(-1, 1, \ldots, 1)$.
2. $$O_2 = \left\{ \begin{bmatrix} a & -b \\ b & a \end{bmatrix} : a^2 + b^2 = 1 \right\} \sqcup \left\{ \begin{bmatrix} a & b \\ b & -a \end{bmatrix} : a^2 + b^2 = 1 \right\}.$$

Theorem 7.2 *continued*

The first set is SO_2: rotation by θ, where $(a, b) = (\cos\theta, \sin\theta)$. The second set consists of reflections in the bisector of the rays $\mathbb{R}_{\geq 0}e_1$ and $\mathbb{R}_{\geq 0}(a, b)^T$, that is, in the line at angle $\theta/2$.



det = 1: rotation by θ det = -1: reflection

Here $(a, b) = (\cos\theta, \sin\theta)$. In both cases $Ae_1 = (a, b)^T$; the second column is $(-b, a)^T$ for a rotation and $(b, -a)^T$ for a reflection.

Proof (class). (a) Every $A \in O_n$ has $\det A = \pm 1$. If $\det A = 1$, $A \in SO_n$. If $\det A = -1$, then $M^{-1}A$ is orthogonal with determinant $(-1)(-1) = 1$, so $A = M(M^{-1}A) \in M \cdot SO_n$. The two sets are disjoint because their determinants differ.

(b) By proposition 7.1(c), the columns of $A \in O_2$ are orthonormal. The first column $(a, b)^T$ is a unit vector, so $a^2 + b^2 = 1$. The second is a unit vector perpendicular to $(a, b)^T$, and in $\mathbb{R}^2$ the only such vectors are $\pm(-b, a)^T$. The sign $+$ gives determinant $a^2 + b^2 = 1$, the sign $-$ gives -1. For the geometric description of the second set, write $(a, b) = (\cos\theta, \sin\theta)$ and check that $\begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$ fixes $(\cos\frac{\theta}{2}, \sin\frac{\theta}{2})^T$ and negates the perpendicular vector $(-\sin\frac{\theta}{2}, \cos\frac{\theta}{2})^T$ (use the angle-difference formulas) (*Ex*). □

Another proof of (b) by eigenvalues. If $A \in O_2$ has $\det A = -1$, then $p_A(x) = x^2 - (\operatorname{tr} A)x - 1$. Its discriminant $(\operatorname{tr} A)^2 + 4$ is positive, so the roots are real; real eigenvalues of an orthogonal matrix are ± 1, and their product is $\det A = -1$, so they are 1 and -1. Eigenvectors for 1 and -1 are orthogonal ($\langle u, w\rangle = \langle Au, Aw\rangle = -\langle u, w\rangle$), so A is the reflection in the 1-eigenline (*Ex*). If $\det A = 1$, write A in terms of its first column (*Ex*). □

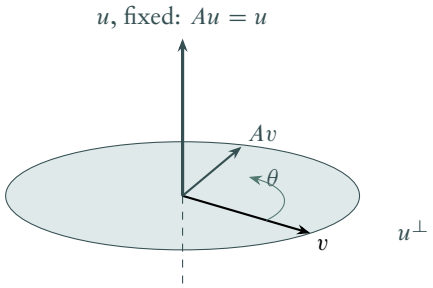
7.3 Euler's theorem on rotations of $\mathbb{R}^3$

Definition 7.3

A *rotation* of $\mathbb{R}^3$ by angle θ about the axis $\mathbb{R}u$ (u a unit vector) is the linear map that fixes u and acts on the plane $u^\perp$ as rotation by θ. In an ON basis v, w, u

Definition 7.3 *continued*

with $\det[v\ w\ u] = 1$ its matrix is $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$.



Euler, Oct 21 **Theorem 7.3**

SO_3 is exactly the set of rotations of $\mathbb{R}^3$. That is, every $A \in SO_3$ fixes a unit vector u and acts on $u^\perp$ as a rotation, and every rotation lies in SO_3.

Proof 1: $\det(A - I) = 0$. For $A \in SO_3$,

$$\begin{aligned} \det(A - I) &= \det\big(A^T\big)\det(A - I) = \det\big(A^T A - A^T\big) = \det\big(I - A^T\big) \\ &= \det\big((I - A)^T\big) = \det(I - A) = (-1)^3 \det(A - I), \end{aligned}$$

using $\det A^T = \det A = 1$, $A^T A = I$, $\det M^T = \det M$, and $\det(-M) = (-1)^3 \det M$ for 3×3 matrices. So $\det(A - I) = 0$, and 1 is an eigenvalue.

The axis. Let u be a unit vector with $Au = u$. The line $\mathbb{R}u$ is A-invariant, so by proposition 7.1(f) the plane $u^\perp$ is A-invariant too.

The plane. Choose an ON basis v, w of $u^\perp$, ordered so that $\det[v\ w\ u] = 1$ (swap v, w if needed). In the ON basis v, w, u: the third column is $(0, 0, 1)^T$ since $Au = u$, and the first two columns have third entry 0 since $Av, Aw \in u^\perp$. So the matrix is $\begin{bmatrix} B & 0 \\ 0 & 1 \end{bmatrix}$, where B is the matrix of $A|_{u^\perp}$. This matrix is orthogonal (ON basis), so $B \in O_2$, and $\det B = \det A = 1$. By theorem 7.2(b), B is a rotation by some θ.

Conversely, a rotation has matrix $QR_\theta Q^{-1}$ in the standard basis, with $Q = [v\ w\ u] \in SO_3$ and R_θ as in the definition; this is a product of orthogonal matrices with determinant 1. □

Proof 2: eigenvalues. p_A is a real cubic, so it has a real root (intermediate value theorem), and every complex root has absolute value 1 by proposition 7.1(e) applied to A as a unitary matrix on $\mathbb{C}^3$. The non-real roots come in conjugate pairs $\mu, \bar\mu$ with $\mu\bar\mu = 1$. The product of all three roots is $\det A = 1$. Case analysis: if there are non-real roots $\mu, \bar\mu$, the real root is 1. If all roots are real, they lie in $\{1, -1\}$ with product 1, so they are $1, 1, 1$ or $1, -1, -1$. In every case 1 is a root (*Ex*). Then continue as in Proof 1. □

Proof 3: Artin 5.1.5, finding the axis. If $A \in SO_3$ is a rotation by $\theta \neq 0, \pi$, then $M = A - A^T$ is skew-symmetric of rank 2, and a nonzero vector in $\ker M$ is an eigenvector of A with eigenvalue 1. Explicitly, if $M = \begin{bmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{bmatrix}$, then $(a, b, c)^T$ spans the axis (*Ex*). Also $\operatorname{tr} A = 1 + 2\cos\theta$, which gives the angle up to sign (*Ex*). □

Corollary 7.4.

1. *The composition of two rotations of $\mathbb{R}^3$ is a rotation (since SO_3 is a group); this is not obvious geometrically (Artin 5.M.8).*
2. *If $A \in O_3$ and $\det A = -1$, then $-A \in SO_3$, so A is a rotation followed by $-I$; equivalently A is a rotation about an axis $\mathbb{R}u$ composed with the reflection in $u^\perp$, and -1 is an eigenvalue of A (Ex).*

Example 7.1 **Reflection in a hyperplane**

For a unit vector $w \in \mathbb{R}^n$, let $P = I - 2ww^T$. Then $Pw = -w$ and $Pu = u$ for $u \perp w$. Also $P^T = P$ and $P^2 = I$, so $P \in O_n$ with $\det P = -1$. The 2025 Test 2 Q4 asks you to derive this from $v = aw + z$ with $z \perp w$.

Exercises

From the board and the tutorials

7.3.1 *Oct 20, 21 boards* **[1]** Show that a linear T with $\|Tv\| = \|v\|$ for all v is orthogonal (or unitary), using both polarization identities.

7.3.2 *Oct 21 board* **[1]** Show $O_n = SO_n \sqcup M\, SO_n$, classify O_2, and check the reflection claim.

7.3.3 *Oct 21 board* **[1]** Prove Euler's theorem. What does "rotation" mean in $\mathbb{R}^3$?

7.3.4 *Oct 22 board* **[1]** If W is T-invariant for orthogonal or unitary T, then $W^\perp$ is T-invariant. (Use that T is invertible.)

7.3.5 *Oct 22 board* **[2]** Show that, with respect to an ON basis, T is unitary if and only if $A^*A = I$.

From the books

7.3.6 *Artin 5.1.1* **[1]** Find the matrices of these rotations of $\mathbb{R}^3$: (a) angle θ, axis e_2; (b) angle $2\pi/3$, axis through $(1, 1, 1)^T$; (c) angle $\pi/2$, axis through $(1, 1, 0)^T$.

7.3.7 *Artin 5.1.2, 5.1.3, 5.1.4* **[2]** (a) What are the complex eigenvalues of a rotation of $\mathbb{R}^3$ by θ about a pole u? (b) Is O_n isomorphic to the product group $SO_n \times \{\pm I\}$? (The answer depends on n.) (c) Describe geometrically the action of a 3×3 orthogonal matrix with determinant -1.

7.3.8 *Artin 5.1.5* **[2]** Let $A \in SO_3$ have angle of rotation different from 0 and π, and $M = A - A^T$. (a) Show M has rank 2 and a nonzero vector of $\ker M$ is an eigenvector of A with eigenvalue 1. (b) Find such an eigenvector explicitly in terms of the entries of A.

7.3.9 *Artin 8.5.3* **[1]** Let $w \in \mathbb{R}^n$ be a unit vector, $U = w^\perp$, and $r_w(cw+u) = -cw+u$ for $u \in U$. (a) Prove $P = I - 2ww^T$ is orthogonal. (b) Prove that multiplication by P is r_w. (c) For u, v of equal length, find w with $Pu = v$.

7.3.10 *Artin 8.5.5, 8.5.6* **[2]** (a) Let P be unitary with eigenvectors X_1, X_2 for distinct eigenvalues. Prove $X_1 \perp X_2$. (b) Which complex numbers can be eigenvalues of a unitary matrix?

7.3.11 *Artin 8.M.11* **[3]** Prove that a real $n\times n$ matrix A is the orthogonal projection to its image if and only if $A^2 = A = A^T A$.

7.3.12 *Axler 7D.1, 7D.3* **[3]** (a) If $\dim V \geq 2$, prove that S preserves inner products if and only if it maps every orthonormal list of length 2 to an orthonormal list. (b) Show that products and inverses of unitary operators are unitary.

7.3.13 *Axler 7D.8, 7D.12, 7D.13* **[3]** (a) Prove or disprove: if $\|Se_k\| = 1$ for every vector of some ON basis, then S is unitary. (b) Prove or disprove: if S is invertible and $\|S^{-1}v\| = \|Sv\|$ for all v, then S is unitary. (c) Explain why the columns of a square complex matrix are orthonormal if and only if its rows are.

7.3.14 *Axler 7D.11, 7D.14* **[3]** (a) Prove that S is unitary if and only if S maps the closed unit ball onto itself. (b) Let $\|v\| = 1$, $b \in \mathbb{F}$ and $\dim V \geq 2$. Prove there is a unitary S with $\langle Sv, v\rangle = b$ if and only if $|b| \leq 1$.

7.3.15 *Halmos §74.3, §74.5* **[2]** (a) Find a 3×3 orthogonal matrix whose first row is a multiple of $(1,1,1)$. (b) Prove that a triangular orthogonal (or unitary) matrix is diagonal.

7.3.16 *Halmos §74.4* **[3]** For a real square matrix A, prove that any two of $A^T = A$, $A^T A = I$, $A^2 = I$ imply the third.

Harder or unusual

7.3.17 *Axler 7D.2* **[4]** Let $T \neq 0$ preserve orthogonality: $\langle u, v\rangle = 0 \Rightarrow \langle Tu, Tv\rangle = 0$. Prove that T is a scalar multiple of an isometry. (Hint: compare $\|Tu\|$ and $\|Tv\|$ for orthonormal u, v using $u + v \perp u - v$.)

7.3.18 *Halmos §74.8* **[4]** Let U be a map (not assumed linear) of an inner product space onto itself with $\langle Ux, Uy\rangle = \langle x, y\rangle$ for all x, y. Prove that U is linear. (Hint: expand $\|U(x + y) - Ux - Uy\|^2$.)

7.3.19 *Halmos §74.6* **[4]** Prove that there is an isometry U with $Ux_i = y_i$ $(i = 1, \ldots, k)$ if and only if the Gram matrices $(\langle x_i, x_j\rangle)$ and $(\langle y_i, y_j\rangle)$ are equal.

7.3.20 *Halmos §74.7* **[4]** (Cayley transform.) Let A be real with $A^T = -A$. Prove that $A - I$ is invertible and that $U = (A + I)(A - I)^{-1}$ is orthogonal with $U - I$ invertible. Conversely, if U is orthogonal and $U - I$ is invertible, then $A = (U + I)(U - I)^{-1}$ satisfies $A^T = -A$.

7.3.21 *Artin 5.M.1* **[3]** Find all orthogonal matrices with integer entries. How many are there in O_n?

7.3.22 *Artin 8.M.6* **[3]** What is wrong with this argument: "Let P be real orthogonal and $PX = \lambda X$. Then $X^T P^T X = (PX)^T X = \lambda X^T X$, and also $X^T P^T X = X^T P^{-1} X = \lambda^{-1} X^T X$, so $\lambda = \pm 1$." State and prove a correct theorem based on the error.

7.3.23 **[4]** Prove that every element of SO_3 is a product of two reflections in planes through 0, and every element of O_3 is a product of at most three. More generally, show by induction that every element of O_n is a product of at most n reflections $I - 2ww^T$. (Hint: send e_1 to Ae_1 by one reflection, using Artin 8.5.3(c).)

7.3.24 *Axler 7D.19* **[4]** (Discrete Fourier transform.) On $\mathbb{C}^n$ define F by

$$F(z)_j = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} z_m e^{-2\pi i j m/n} \qquad (j = 0, \ldots, n-1).$$

Show that F is unitary and $F^4 = I$.

7.3.25 *Artin 8.M.12(a), beyond the syllabus* **[4]** Let A be real orthogonal and $X = R + iS$ a complex eigenvector with non-real eigenvalue λ (R, S real). Prove $X^T X = 0$, that $W = \operatorname{span}(R, S)$ is a 2-dimensional A-invariant plane, and that $A|_W$ is a rotation.

Part 3



The Test

CHAPTER 8	Writing and Standard Methods	63
CHAPTER 9	Mock Test	66
CHAPTER A	Fields and Counting over $\mathbb{F}_p$	68

8 Writing and Standard Methods

8.1 Order of work

The rank-1 exercises, in the recommended order, together with the past papers. Each line gives the chapter and the source shown next to the exercise number.

1. Chapter 4: Step 3 (Oct 14 board). This proof is required.
2. Chapter 1: Tutorial 6 Q1(a),(b); Q1(c); Q5(a); Q10 first part.
3. Chapter 1: Axler 5D.21 (powers by diagonalizing); Axler 5D.6, 5D.9, 5D.10 (similarity without Jordan form).
4. Chapter 2: Artin 4.4.6, 4.6.8; Axler 8A.1, 8A.2.
5. Chapter 3: Tutorial 6 Q7; HW6 Q4; Axler 5D.1.
6. Chapter 3: Artin 4.5.9, 4.5.10, 4.3.4; Artin 5.2.1, 5.2.2; Artin 5.2.3, 5.2.4 (operators on matrix spaces).
7. Chapter 4: HW6 Q6, Q7; Artin 4.7.1–4.7.2, 4.7.3–4.7.5, 4.7.7–4.7.9.
8. Chapter 5: Oct 15 board (Gram–Schmidt on polynomials); Oct 15/22 boards ($W^{\perp}$); Oct 15/20 boards (Cauchy–Schwarz); Artin 8.4.6, 8.4.7; Artin 8.4.15, 8.4.16.
9. Chapter 6: Oct 22 boards (axioms; Cauchy–Schwarz over $\mathbb{C}$); Artin 8.4.13(a), Axler 6A.29 (the inner product $\operatorname{tr}(A^*B)$).
10. Chapter 7: Oct 20–22 boards (norm-preserving maps, O_n and O_2, Euler, invariance of $W^{\perp}$); Artin 5.1.1; Artin 8.5.3.
11. The 2025 Test 2 (timed), then the mock test in chapter 9.

Then rank-2 exercises, chapter by chapter, starting with Chapters 4 and 7.

8.2 Rules for the answers

The 2025 Test 2 and Endsem opened with these instructions, and the same instructor sets this test.

- "Explain clearly." Every claim needs a reason. When you use a result from class, name it (for example "by Cayley–Hamilton", "since eigenspaces for distinct eigenvalues are independent").
- "Do not use the Jordan form unless the question mentions it." Use eigenspace dimensions, $\dim\ker(A-\lambda I)^i$, characteristic and minimal polynomials, and triangular form instead.
- "Unless the question asks you to prove something standard." If it does, the standard result cannot be quoted.
- In 2025, Q0 gave 0 credit for formulas guessed and proved by induction. When a question says "using linear algebra", the method is marked, not only the answer.
- Separate file for each problem, uploaded to Moodle.

From the Test 1 script:

- Say which basis you use and in what order before writing a matrix.
- Keep the types straight: a vector of $\mathbb{F}^n$, a matrix in M_n, and a coordinate vector of length n^2 are different objects. Write $X = [x \mid 0 \mid \cdots \mid 0]$ when a vector has to become a matrix.
- Write one clean final version for each problem. Do the rough work separately; do not upload narration or crossed-out attempts.

Checks that cost a minute each: $P^{-1}AP = D$ (or $AP = PD$) for a diagonalization; $A^TA = I$ for an orthogonal matrix; $\det = +1$ for a rotation; the projection of v minus v is perpendicular to the subspace; a Jordan form has the right total size and the right $\dim\ker(A-\lambda I)^i$.

8.3 Standard computations

1. *Powers M^n.* If $M = PDP^{-1}$, then $M^n = PD^nP^{-1}$; multiply out to get each entry as a combination of λ_i^n. If a 2×2 matrix has $p_M = (x-\lambda)^2$, write $M = \lambda I + N$ with $N = M - \lambda I$; Cayley–Hamilton gives $N^2 = 0$, and since λI and N commute, $M^n = \lambda^n I + n\lambda^{n-1}N$.
2. *Diagonalizable or not.* Factor p_T; for each root compare $\dim\ker(\lambda I - T) = n - \operatorname{rank}(\lambda I - T)$ with the multiplicity (corollary 1.3). Or check whether m_T has distinct roots.
3. *Jordan type* (only when allowed). For each λ, compute $d_i = \dim\ker(T-\lambda I)^i$ until it reaches the multiplicity; the number of blocks of size $\geq i$ is $d_i - d_{i-1}$ (theorem 4.5).
4. *Similar or not.* Similar matrices have equal characteristic polynomials and equal $\dim\ker(A-\lambda I)^i$ for all λ, i. With Jordan form, these numbers decide similarity.

5. *Orthonormal basis of a plane $ax+by+cz=0$ in $\mathbb{R}^3$.* The normal is $n=(a,b,c)$. Pick u in the plane, then $v \perp u$ in the plane (solve two linear equations, or take the cross product $n \times u$), and normalize. Then $[u\ v\ n]$ with normalized columns is orthogonal.

6. *Rotation by θ about a unit axis u in $\mathbb{R}^3$.* Take an ON basis v, w, u with $\det[v\ w\ u] = 1$, let $Q = [v\ w\ u]$, and

$$R = Q \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} Q^T.$$

7. *Reflections and projections.* Reflection in $w^\perp$: $I - \frac{2}{\|w\|^2} ww^T$. Projection onto $w^\perp$: $v \mapsto v - \frac{\langle w,v\rangle}{\|w\|^2} w$. Projection onto W with ON basis $e_1, \ldots, e_k$: $\sum \langle e_i, v\rangle\, e_i$.

8. *Maximize a linear function on the unit sphere.* $\langle a, x\rangle \le \|a\|\|x\| = \|a\|$ by Cauchy–Schwarz, with equality if and only if $x = a/\|a\|$.

9 Mock Test

A practice paper in the style of the 2025 Test 2, written for these notes. Time: 4 hours. Explain clearly. Do not use the Jordan form unless the question mentions it. You may use any result from class; name it. No solutions are included.

0. Let $M = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ and $M^n = \begin{bmatrix} a(n) & b(n) \\ c(n) & d(n) \end{bmatrix}$ for $n \geq 1$. Using linear algebra (not guessing and induction), find formulas for a, b, c, d involving only arithmetic on scalars.

1. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ and let $T : M_2(\mathbb{R}) \to M_2(\mathbb{R})$ be $T(X) = AX - XA$.
   (a) Find the eigenvalues of T and decide whether T is diagonalizable.
   (b) Find $\ker T$ and the rank of T.
   (c) Find $\det(T + I)$ and $\operatorname{tr}(T + I)$.

2. Let S be the set of 6×6 matrices A over $\mathbb{F}_5$ with $p_A(x) = x^4(x-1)^2$, $\dim\ker A = 2$ and $\dim\ker A^2 = 3$.
   (a) Show that every $A \in S$ has rank $A^3 = 2$.
   (b) Using the Jordan form, list the Jordan forms of matrices in S. How many similarity classes does S contain?
   (c) Using the Jordan form, decide whether all $A \in S$ with $\dim\ker(A - I) = 1$ are similar to each other.

3. Let N be a nilpotent 7×7 complex matrix.
   (a) If $\operatorname{rank} N = 4$, $\operatorname{rank} N^2 = 2$ and $\operatorname{rank} N^3 = 1$, find the Jordan form of N.
   (b) How many similarity classes of nilpotent 7×7 matrices have $\operatorname{rank} N = 4$?
   (c) Can $\operatorname{rank} N = 3$ and $\operatorname{rank} N^2 = 0$ both hold? Explain using the dimensions of kernels.

4. Let $U = \{(x, y, z)^T : x + 2y + 2z = 0\} \subseteq \mathbb{R}^3$.
   (a) Find an orthogonal matrix $Q = [u\ v\ w]$ with $\det Q = 1$ such that u, v is a basis of U.
   (b) Write the rotation of $\mathbb{R}^3$ by $\pi/2$ about the axis $\mathbb{R}w$ as a product of explicit matrices.

(c) Find the orthogonal projection of $(1,1,1)^T$ onto U, and the reflection of $(1,1,1)^T$ in U.

5. In $V =$ the real polynomials of degree at most 2, with $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$:
   (a) Apply Gram–Schmidt to $1, x, x^2$.
   (b) Let D be differentiation on V. Find the matrix of D in the basis from (a). Why is it upper triangular?

6. Short questions. Justify each answer.
   (a) If $A \in O_3$ and $\det A = -1$, show that -1 is an eigenvalue of A.
   (b) Prove or disprove: every operator P on $\mathbb{R}^2$ (dot product) with $P^2 = P$ is an orthogonal projection.
   (c) Find the maximum of $x + 2y - 2z$ on the unit sphere of $\mathbb{R}^3$ and where it is attained.
   (d) Let P be a unitary matrix with eigenvectors X_1, X_2 for distinct eigenvalues. Show $X_1 \perp X_2$.
   (e) Let A be 3×3 with $p_A(x) = x^3 - 2x + 1$. Express A^{-1} as a polynomial in A.
   (f) Let T be unitary on $\mathbb{C}^n$ and W a T-invariant subspace. Show that $W^\perp$ is T-invariant.
   (g) Give an example of two 4×4 nilpotent matrices with the same rank that are not similar.

Fields and Counting over $\mathbb{F}_p$



The Sep 28 lecture (before eigenanalysis) also counts for the test.

Definition A.1

A *field* is a set with $+, -, \times, \nabla\cdot$ (by nonzero elements) that obey the usual rules. Examples: $\mathbb{Q} \subset \mathbb{Q}[\sqrt{n}] \subset \mathbb{R} \subset \mathbb{C}$, $\mathbb{Q}[i]$; the rational functions $\mathbb{F}(x)$; and $\mathbb{F}_p = \{\bar{0}, \ldots, \overline{p-1}\}$, the integers mod a prime p.

Proposition A.1. *$\bar{k}$ is invertible mod n if and only if* $\gcd(k, n) = 1$. *So $\mathbb{Z}/n$ is a field if and only if n is prime.*

Proof outline. If $\gcd(k, n) = 1$, Bézout gives $ak + bn = 1$, so $\bar{a}\bar{k} = \bar{1}$. If $\gcd(k, n) = d > 1$ and $\bar{k}\bar{x} = \bar{1}$, multiply by $\overline{n/d}$ to get $\overline{n/d} = \bar{0}$, a contradiction (*Ex*). □

Everything in linear algebra so far works over any field. Counting over $\mathbb{F}_q$ ($q = p$):

- $|GL_n(\mathbb{F}_q)| = (q^n - 1)(q^n - q)\cdots(q^n - q^{n-1})$: choose the columns one at a time, each outside the span of the previous ones. For example $|GL_2(\mathbb{F}_p)| = p(p+1)(p-1)^2$.
- $|SL_n(\mathbb{F}_q)| = |GL_n(\mathbb{F}_q)|/(q-1)$, since det: $GL_n \to \mathbb{F}_q^\times$ is onto and all fibres have the same size (*Ex*).
- The number of k-dimensional subspaces of $\mathbb{F}_q^n$ is

$$\frac{(q^n - 1)(q^n - q)\cdots(q^n - q^{k-1})}{(q^k - 1)(q^k - q)\cdots(q^k - q^{k-1})},$$

  the number of independent k-lists divided by the number of bases of one k-dimensional space.

Exercises

A.0.1 [2] Tutorial 6 Q3: $|GL_n(\mathbb{F}_p)|$, $|SL_n(\mathbb{F}_p)|$, and the number of subspaces of each dimension in $\mathbb{F}_p^3$.

A.0.2 [4] Tutorial 6 Q2 (SMMC 2023).